

Signals and Systems

Previous Year Questions • 1992–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Signals and Systems (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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PrepFusion

UNIT

1

Basics of Signals

Topics

1. Basics of Signals and Mathematical Foundation
2. Step and Rectangular Signal
3. Ramp and Triangular Signal
4. Impulse Signal
5. Miscellaneous Signals
6. Complex Exponential Signal
7. Periodicity of a Signal
8. Orthogonality of Signals
9. Energy, Average and Power of a Signal
10. Even, Odd, Conjugate Symmetry Signals
11. Discrete-Time Signals and Their Types
12. Classification of Discrete-Time Signals

Unit Snapshot*

Stream: EE	**Questions:** 28	**MCQ:** 22	**MSQ:** 1	**NAT:** 5
1M: 19	**2M:** 9	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 1M

Let $x_c(t)$ be any continuous-time periodic signal with period T. It is sampled uniformly with a sampling period T_S where $T_S \neq T$, resulting in the discrete sequence

$$x[n] = x_c(nT_S), \text{ where } n \text{ is an integer.}$$

Which one of the following statements is correct about $x[n]$?

A. $x[n]$ will always be periodic with period T/T_S for all values of T/T_S

B. $x[n]$ will always be periodic with period 1 for all values of T/T_S

C. $x[n]$ will never be periodic

D. $x[n]$ will be periodic if and only if T/T_S is a rational number

Q1.25.1 MCQ 2025 1M

Consider a continuous-time signal

$$x(t) = -t^2\{u(t+4) - u(t-4)\}$$

where $u(t)$ is the continuous-time unit step function. Let $\delta(t)$ be the continuous-time unit impulse function. The value of

$$\int_{-\infty}^{\infty} x(t)\,\delta(t+3)\,dt$$

is

A. −9 **B.** 9 **C.** 3 **D.** −3

Q1.25.2 NAT 2025 1M

A continuous time periodic signal $x(t)$ is

$$x(t) = 1 + 2\cos 2\pi t + 2\cos 4\pi t + 2\cos 6\pi t$$

If T is the period of $x(t)$, then

$$\frac{1}{T}\int_0^T |x(t)|^2\,dt = _____$$

(Round off to the nearest integer.)

Q1.24.1 NAT 2024 2M

If the energy of a continuous-time signal $x(t)$ is E and the energy of the signal $2x(2t-1)$ is cE, then c is _____ (rounded off to 1 decimal place).

Q1.23.1 NAT 2023 2M

The period of the discrete time signal $x[n]$ described by the equation below is $N =$ _____ (Round off to the nearest integer).

$$x[n] = 1 + 3\sin\left(\frac{15\pi}{8}n + \frac{3\pi}{4}\right) - 5\sin\left(\frac{\pi}{3}n - \frac{\pi}{4}\right)$$

Q1.20.1 MSQ 2020 1M

Which of the following is true for all possible non-zero choices of integers $m, n;\ m \neq n$, or all possible non-zero choices of real numbers $p, q;\ p \neq q$, as applicable?

☐ **A.** $\lim_{a\to\infty} \frac{1}{2a}\int_{-a}^{a} \sin p\theta \cos q\theta\, d\theta = 0$

☐ **B.** $\frac{1}{2\pi}\int_{-\pi}^{\pi} \sin p\theta \cos q\theta\, d\theta = 0$

☐ **C.** $\frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} \sin p\theta \sin q\theta\, d\theta = 0$

☐ **D.** $\frac{1}{2\pi}\int_{-\pi}^{\pi} \sin m\theta \sin n\theta\, d\theta = 0$

Q1.20.2 MCQ 2020 1M

x_R and x_A are, respectively, the rms and average values of $x(t) = x(t-T)$, and similarly, y_R and y_A are respectively, the rms and average values of $y(t) = kx(t)$. k, T are independent of t. Which of the following is true?

A. $y_A = kx_A;\ y_R = kx_R$

B. $y_A = kx_A;\ y_R \neq kx_R$

C. $y_A \neq kx_A;\ y_R = kx_R$

D. $y_A \neq kx_A;\ y_R \neq kx_R$

Q1.18.1 MCQ 2018 2M

Consider the two continuous-time signals defined below:

$$x_1(t) = \begin{cases} |t|, & -1 \le t \le 1 \\ 0, & \text{Otherwise} \end{cases}$$

$$x_2(t) = \begin{cases} 1-|t|, & -1 \le t \le 1 \\ 0, & \text{Otherwise} \end{cases}$$

These signals are sampled with a sampling period of $T = 0.25$ seconds to obtain discrete-time signals $x_1[n]$ and $x_2[n]$, respectively. Which one of the following statements is true?

A. The energy of $x_1[n]$ is greater than the energy of $x_2[n]$.

B. The energy of $x_2[n]$ is greater than the energy of $x_1[n]$.

C. $x_1[n]$ and $x_2[n]$ have equal energies.

D. Neither $x_1[n]$ nor $x_2[n]$ is a finite-energy signal.

Q1.18.2 MCQ 2018 2M

The signal energy of the continuous time signal

$$x(t) = [(t-1)u(t-1)] - [(t-2)u(t-2)] - [(t-3)u(t-3)] + [(t-4)u(t-4)]$$

A. $\frac{11}{3}$ **B.** $\frac{7}{3}$ **C.** $\frac{1}{3}$ **D.** $\frac{5}{3}$

Q1.17.1 NAT 2017 1M

The mean square value of the given periodic waveform $f(t)$ is _____.

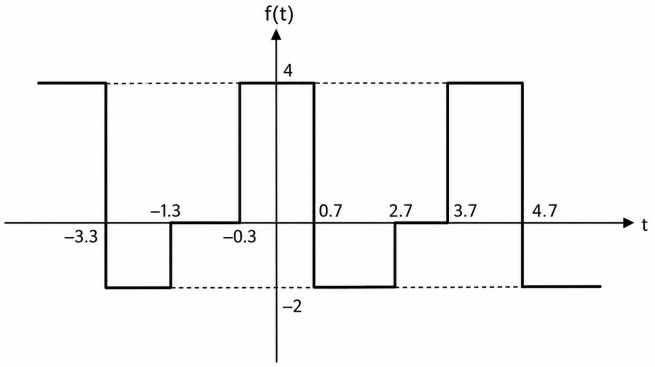


Fig. Q1.17.1

Q1.16.1 MCQ 2016 1M

The value of

$$\int_{-\infty}^{\infty} e^{-t}\,\delta(2t-2)\,dt,$$

where $\delta(t)$ is the Dirac delta function, is

A. $\frac{1}{2e}$ **B.** $\frac{2}{e}$ **C.** $\frac{1}{e^2}$ **D.** $\frac{1}{2e^2}$

Q1.16.2 MCQ 2016 1M

The value of the integral $2\int_{-\infty}^{\infty}\left(\frac{\sin 2\pi t}{\pi t}\right)dt$ is equal to

A. 0 **B.** 0.5 **C.** 1 **D.** 2

Q1.15.1 NAT 2015 1M

A moving average function is given by

$$y(t) = \frac{1}{T}\int_{t-T}^{t} u(\tau)\,d\tau$$

If the input u is a sinusoidal signal of frequency $\frac{1}{2T}$ Hz, then in steady state, the output y will lag u (in degree) by _____.

Q1.14.1 MCQ 2014 2M

The function shown in the figure can be represented as

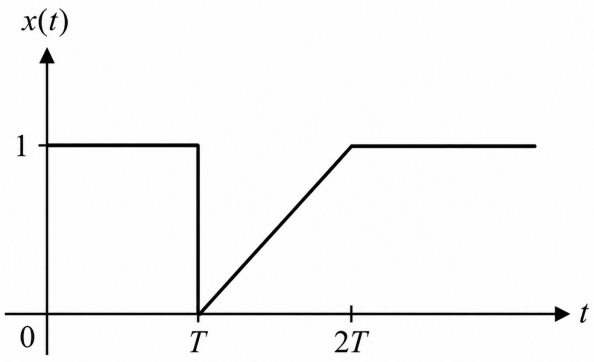


Fig. Q1.14.1

A. $u(t) - u(t-T) + \frac{(t-T)}{T}u(t-T) - \frac{(t-2T)}{T}u(t-2T)$

B. $u(t) + \frac{t}{T}u(t-T) - \frac{t}{T}u(t-2T)$

C. $u(t) - u(t-T) + \frac{(t-T)}{T}u(t) - \frac{(t-2T)}{T}u(t)$

D. $u(t) + \frac{(t-T)}{T}u(t-T) - 2\frac{(t-2T)}{T}u(t-2T)$

Q1.11.1 MCQ 2011 2M

A zero mean random signal is uniformly distributed between limits $-a$ and $+a$ and its mean square value is equal to its variance. Then the RMS value of the signal is

A. $\frac{a}{\sqrt{3}}$ **B.** $\frac{a}{\sqrt{2}}$ **C.** $a\sqrt{2}$ **D.** $a\sqrt{3}$

Q1.10.1 MCQ 2010 1M

The period of the signal

$$x(t) = 8\sin\left(0.8\pi t + \frac{\pi}{4}\right)$$

A. 0.4π s **B.** 0.8π s **C.** 1.25 s **D.** 2.5 s

Q1.08.1 MCQ 2008 2M

Given a sequence $x[n]$ to generate the sequence $y[n] = x[3-4n]$, which one of the following procedures would be correct?

A. First delay $x[n]$ by 3 samples to generate $z_1[n]$, then pick every 4th sample of $z_1[n]$ to generate $z_2[n]$ and then finally time reverse $z_2[n]$ to obtain $y[n]$.

B. First advance $x[n]$ by 3 samples to generate $z_1[n]$, then pick every 4th sample of $z_1[n]$ to generate $z_2[n]$ and then finally time reverse $z_2[n]$ to obtain $y[n]$.

C. First pick every fourth samples of $x[n]$ to generate $v_1[n]$, time-reverse $v_1[n]$ to obtain $v_2[n]$, and finally advance $v_2[n]$ by 3 samples to obtain $y[n]$.

D. First pick every fourth sample of $x[n]$ to generate $v_1[n]$, time-reverse $v_1[n]$ to obtain $v_2[n]$, and finally delay $v_2[n]$ by 3 samples to obtain $y[n]$.

Q1.06.1 MCQ 2006 1M

Which of the following is true?

A. A finite signal is always bounded.

B. A bounded signal always possesses finite energy.

C. A bounded signal is always zero outside the interval $[-t_0, t_0]$ for some t_0.

D. A bounded signal is always finite.

Q1.05.1 MCQ 2005 2M

For the triangular waveform shown in the figure, the RMS value of the voltage is equal to

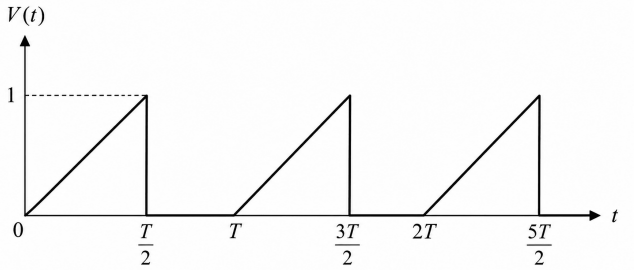


Fig. Q1.05.1

A. $\sqrt{\frac{1}{6}}$ **B.** $\sqrt{\frac{1}{3}}$ **C.** $\frac{1}{3}$ **D.** $\sqrt{\frac{2}{3}}$

Q1.04.1 MCQ 2004 2M

The RMS value of the periodic waveform given in figure is

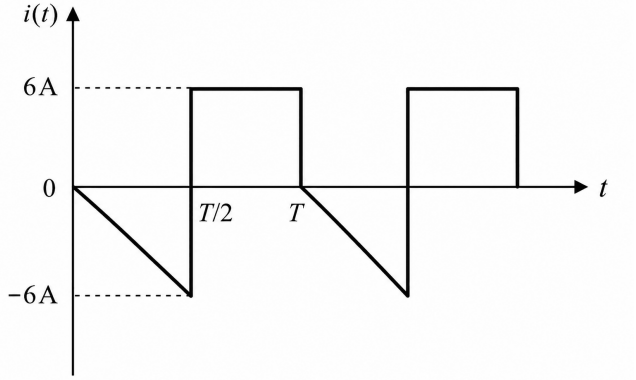


Fig. Q1.04.1

A. $2\sqrt{6}$ A **B.** $6\sqrt{2}$ A **C.** $\sqrt{\frac{4}{3}}$ A **D.** 1.5 A

Q1.04.2 MCQ 2004 1M

The rms value of the resultant current in a wire which carries a dc current of 10 A and a sinusoidal alternating current of peak value of 20 A is

A. 14.1 A **B.** 17.3 A **C.** 22.4 A **D.** 30.0 A

Q1.02.1 MCQ 2002 1M

What is the RMS value of the voltage waveform shown in the figure?

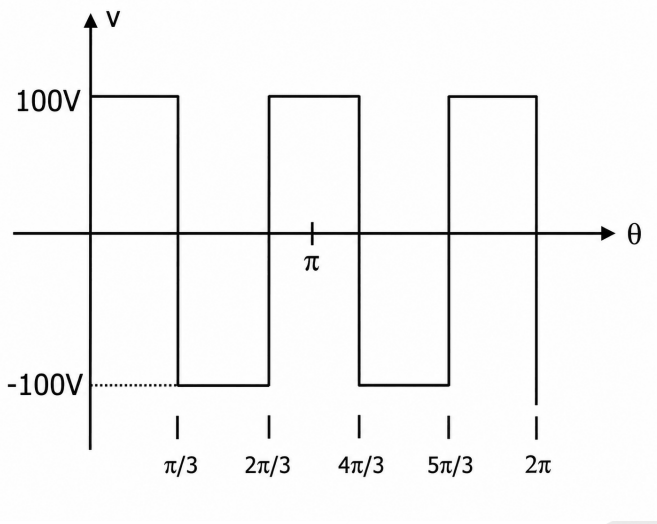


Fig. Q1.02.1

A. $\frac{200}{\pi}$ V **B.** $\left(\frac{100}{\pi}\right)$ V **C.** 200 V **D.** 100 V

Q1.02.2 MCQ 2002 1M

If an a.c. voltage wave is corrupted with an arbitrary number of harmonics, then the overall voltage waveform differs from its fundamental frequency component in terms of

A. only the peak values

B. only the RMS values

C. only the average values

D. all the three measures (peak, RMS and average values)

Q1.02.3 MCQ 2002 1M

A current impulse, $5\delta(t)$, is forced through a capacitor C. The voltage, $V_c(t)$, across the capacitor is given by

A. $5t$ **B.** $5u(t) - C$

C. $\frac{5}{C}$ **D.** $\frac{5u(t)}{C}$

Q1.99.1 MCQ 1999 1M

The RMS value of a half-wave rectified symmetrical square wave current of $2A$ is

A. $\sqrt{2}$ A **B.** 1 A **C.** $\frac{1}{\sqrt{2}}$ A **D.** $\sqrt{3}$ A

Q1.95.1 MCQ 1995 1M

The RMS value of the waveform $s(t)$ shown in fig is

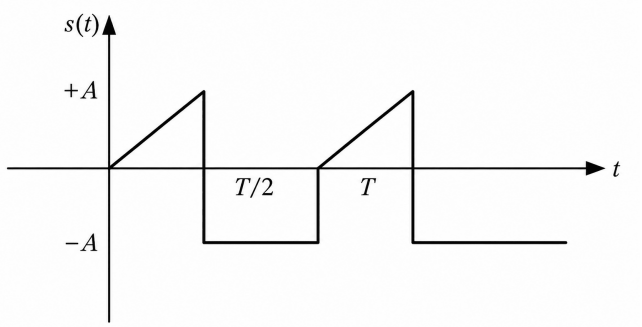


Fig. Q1.95.1

A. $\sqrt{\frac{3}{2}}$ A **B.** $\sqrt{\frac{2}{3}}$ A **C.** $\sqrt{\frac{1}{3}}$ A **D.** $\sqrt{2}$ A

Q1.94.1 MCQ 1994 1M

The value of the integral

$$\int_{-5}^{6} e^{-2t}\delta(t-1)\,dt$$

is equal to

A. 1 **B.** e^{-2} **C.** 0 **D.** e^{2}

Q1.94.2 MCQ 1994 1M

Match the waveforms on the left-hand side with the correct mathematical description listed on the right hand side.

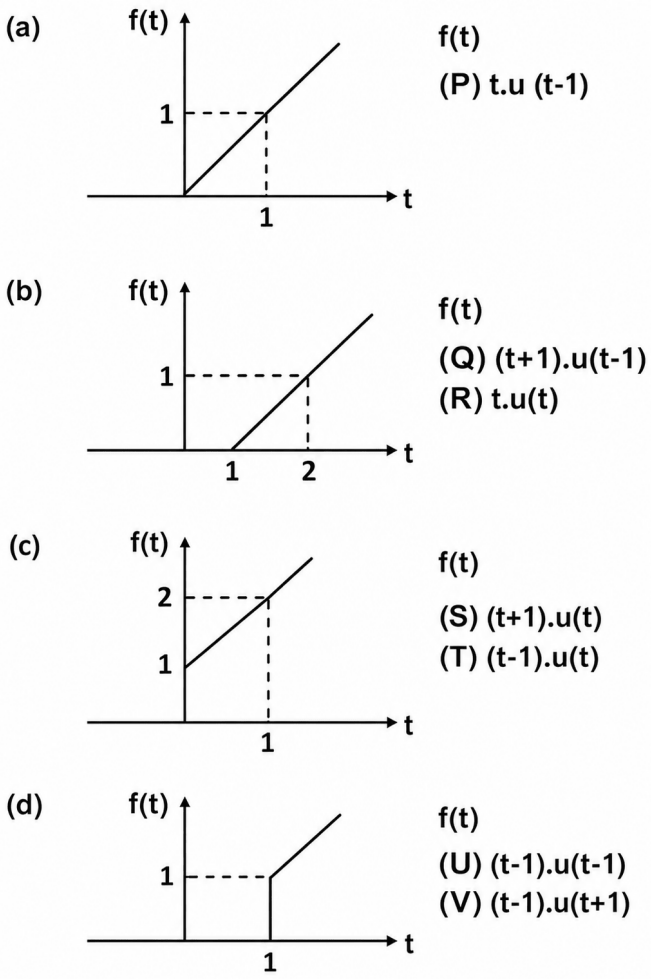


Fig. Q1.94.2

Q1.26.1 D	Q1.25.1 A	Q1.25.2 7
Q1.24.1 2	Q1.23.1 48	Q1.20.1 A,B,D
Q1.20.2 B	Q1.18.1 A	Q1.18.2 D
Q1.17.1 6	Q1.16.1 A	Q1.16.2 D
Q1.15.1 90	Q1.14.1 A	Q1.11.1 A
Q1.10.1 D	Q1.08.1 B	Q1.06.1 MTA
Q1.05.1 A	Q1.04.1 A	Q1.04.2 B
Q1.02.1 D	Q1.02.2 D	Q1.02.3 D
Q1.99.1 A	Q1.95.1 B	Q1.94.1 B
Q1.94.2 B		

A. A-R, B-S, C-U, D-P

B. A-R, B-U, C-S, D-P

C. A-T, B-U, C-S, D-P, D-V

D. A-P, B-T, C-S, D-U

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

UNIT

II

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Unit Snapshot*

Stream: EE	**Questions:** 44	**MCQ:** 40	**MSQ:** 1	**NAT:** 3
1M: 22	**2M:** 21	**Descriptive:** 1	**Avg Weightage** ~3M	**Range:** 0M(2014)–6M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.25.1 MCQ 2025 2M

The continuous-time unit impulse signal is applied as an input to a continuous-time linear time-invariant system S. The output is observed to be the continuous-time unit step signal $u(t)$. Which one of the following statements is true?

A. Every bounded input signal applied to S results in a bounded output signal.

B. It is possible to find a bounded input signal which when applied to S results in an unbounded output signal.

C. On applying any input signal to S, the output signal is always bounded.

D. On applying any input signal to S the output signal is always unbounded.

Q2.25.2 MCQ 2025 2M

Let continuous-time signals $x_1(t)$ and $x_2(t)$ be

$$x_1(t) = \begin{cases} 1, & t \in [0,1] \\ 2-t, & t \in [1,2] \\ 0, & \text{otherwise} \end{cases}$$

$$x_2(t) = \begin{cases} t, & t \in [0,1] \\ 2-t, & t \in [1,2] \\ 0, & \text{otherwise.} \end{cases}$$

Consider the convolution $y(t) = x_1(t) * x_2(t)$. Then

$$\int_{-\infty}^{\infty} y(t)\,dt$$

is

A. 1.5 **B.** 2.5 **C.** 3.5 **D.** 4

Q2.25.3 MCQ 2025 1M

Selected data points of the step response of a stable first-order linear time-invariant (LTI) system are given below. The closest value of the time-constant, in sec, of the system is

Time (sec)	0.6	1.6	2.6	10	∞
Output	0.78	1.65	2.18	2.98	3

A. 1 **B.** 2 **C.** 3 **D.** 4

Q2.25.4 MCQ 2025 1M

Consider a discrete-time linear time-invariant (LTI) system S, where

$$y[n] = S\{x[n]\}$$

Let

$$S\{\delta[n]\} = \begin{cases} 1, & n \in \{0,1,2\} \\ 0, & \text{otherwise} \end{cases}$$

where $\delta[n]$ is the discrete-time unit impulse function. For an input signal $x[n]$, the output $y[n]$ is

A. $x[n] + x[n-1] + x[n-2]$

B. $x[n-1] + x[n] + x[n+1]$

C. $x[n] + x[n+1] + x[n+2]$

D. $x[n+1] + x[n+2] + x[n+3]$

Q2.24.1 MCQ 2024 2M

Consider the discrete-time systems T_1 and T_2 defined as follows:

$$\{T_1 x\}[n] = x[0] + x[1] + \cdots + x[n]$$

$$\{T_2 x\}[n] = x[0] + \frac{1}{2}x[1] + \cdots + \frac{1}{2^n}x[n]$$

Which one of the following statements is true?

A. T_1 and T_2 are BIBO stable.

B. T_1 and T_2 are not BIBO stable.

C. T_1 is BIBO stable but T_2 is not BIBO stable.

D. T_1 is not BIBO stable but T_2 is BIBO stable.

Q2.24.2 MCQ 2024 2M

The input $x(t)$ and the output $y(t)$ of a system are related as

$$y(t) = e^{-t} \int_{-\infty}^{t} e^{\tau} x(\tau)\,d\tau.$$

For all real t, the system is

A. nonlinear.

B. linear and time-invariant.

C. linear but not time-invariant.

D. noncausal.

Q2.24.3 MCQ 2024 1M

Suppose signal $y(t)$ is obtained by the time-reversal of signal $x(t)$, i.e., $y(t) = x(-t)$, $-\infty < t < \infty$. Which one of the following options is always true for the convolution of $x(t)$ and $y(t)$?

A. It is an even signal.

B. It is an odd signal.

C. It is a causal signal.

D. It is an anti-causal signal.

Q2.23.1 MSQ 2023 1M

Which of the following statement(s) is/are true?

☐ **A.** If an LTI system is causal, it is stable.

☐ **B.** A discrete-time LTI system is causal if and only if its response to a step input $u[n]$ is 0 for $n < 0$.

☐ **C.** If a discrete-time LTI system has an impulse response $h[n]$ of finite duration, the system is stable.

☐ **D.** If the impulse response $0 < |h[n]| < 1$ for all n, then the LTI system is stable.

Q2.23.2 NAT 2023 1M

For the signals $x(t)$ and $y(t)$ shown in the figure, $z(t) = x(t) * y(t)$ is maximum at $t = T_1$. Then T_1 in seconds is _____ (Round off to the nearest integer).

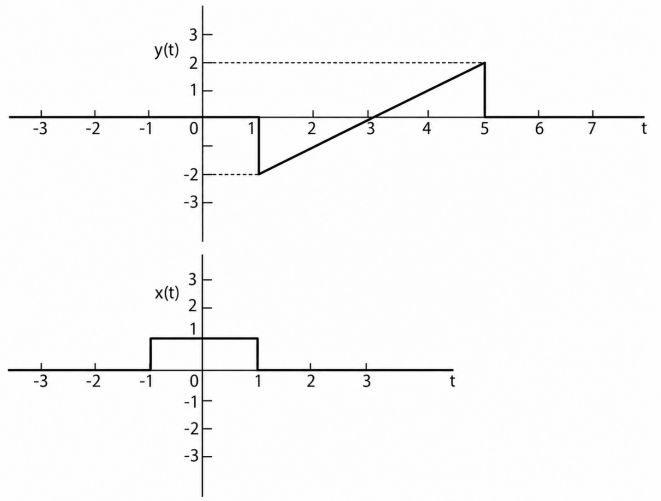


Fig. Q2.23.2

Q2.22.1 MCQ 2022 2M

Consider the system as shown below.

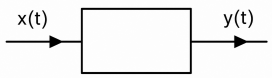


Fig. Q2.22.1

where $y(t) = x(e^t)$. The system is

A. Linear and causal.

B. Linear and non-causal.

C. Non-linear and causal.

D. Non-linear and non-causal.

Q2.22.2 MCQ 2022 2M

Let a causal LTI system be governed by the following differential equation

$$y(t) + \frac{1}{4}\frac{dy(t)}{dt} = 2x(t),$$

where $x(t)$ and $y(t)$ are the input and output respectively. Its impulse response is

A. $2e^{4t}u(t)$ **B.** $2e^{-4t}u(t)$

C. $8e^{-\frac{1}{4}t}u(t)$ **D.** $8e^{-4t}u(t)$

Q2.21.1 MCQ 2021 1M

Two discrete-time linear time-invariant systems with impulse responses

$$h_1[n] = \delta[n-1] + \delta[n+1]$$

and

$$h_2[n] = \delta[n] + \delta[n-1]$$

are connected in cascade, where $\delta[n]$ is the Kronecker delta. The impulse response of the cascaded system is

A. $\delta[n-2] + \delta[n+1]$

B. $\delta[n-1]\delta[n] + \delta[n+1]\delta[n-1]$

C. $\delta[n-2] + \delta[n-1] + \delta[n] + \delta[n+1]$

D. $\delta[n]\delta[n-1] + \delta[n-2]\delta[n+1]$

Q2.21.2 MCQ 2021 1M

If the input $x(t)$ and output $y(t)$ of a system are related as

$$y(t) = \max(0, x(t)),$$

then the system is

A. Linear and time-variant.

B. Linear and time-invariant.

C. Non-linear and time-variant.

D. Non-linear and time-invariant.

Q2.20.1 MCQ 2020 2M

Which of the following options is true for a linear time-invariant discrete-time system that obeys the difference equation:

$$y[n] - ay[n-1] = bx[n] - bx[n-1].$$

A. $y[n]$ is unaffected by the values of $x[n-k]$; $k > 2$.

B. The system is necessarily causal.

C. The system impulse response is non-zero at infinitely many instants.

D. When $x[n] = 0,\ n < 0$, the function $y[n];\ n > 0$ is solely determined by the function $x[n]$.

Q2.20.2 MCQ 2020 2M

Suppose for input $x(t)$ a linear time-invariant system with impulse response $h(t)$ produces output $y(t)$, so that

$$x(t) * h(t) = y(t).$$

Further, if

$$|x(t)| * |h(t)| = z(t),$$

which of the following statements is true?

A. For all $t \in (-\infty, \infty)$, $z(t) \leq y(t)$.

B. For some but not all $t \in (-\infty, \infty)$, $z(t) \leq y(t)$.

C. For all $t \in (-\infty, \infty)$, $z(t) \geq y(t)$.

D. For some but not all $t \in (-\infty, \infty)$, $z(t) \geq y(t)$.

Q2.20.3 MCQ 2020 1M

Consider a linear time-invariant system whose input $r(t)$ and output $y(t)$ are related by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + 4y(t) = 6r(t).$$

The poles of this system are at

A. $+2j, -2j$ **B.** $+2, -2$

C. $+4, -4$ **D.** $+4j, -4j$

Q2.19.1 MCQ 2019 1M

The Symbols, a and T, represent positive quantities, and $u(t)$ is the unit step function. Which one of the following impulse responses is not the output of a causal linear time-invariant system?

A. $e^{-a(t-T)}u(t)$ **B.** $e^{-a(t+T)}u(t)$

C. $e^{at}u(t)$ **D.** $1 + e^{-at}u(t)$

Q2.18.1 MCQ 2018 1M

A continuous-time input signal $x(t)$ is an Eigen function of an LTI system, if the output is

A. $kx(t)$, where k is an Eigen value.

B. $ke^{j\omega t}x(t)$, where k is an Eigen function and $e^{j\omega t}$ is a complex exponential signal.

C. $\overline{x}(t)e^{j\omega t}$, where $e^{j\omega t}$ is a complex exponential signal.

D. $kH(\omega)$, where k is an Eigen function and $H(\omega)$ is a frequency response of the system.

Q2.17.1 MCQ 2017 1M

Consider the system with following input-output relation

$$y[n] = (1 + (-1)^n)\, x[n].$$

Where $x[n]$ is the input and $y[n]$ is the output. The system is

A. Invertible and time invariant.

B. Invertible and time varying.

C. Non-invertible and time invariant.

D. Non-invertible and time varying.

Q2.17.2 NAT 2017 2M

Consider a causal and stable LTI system with rational transfer function $H(z)$, whose corresponding impulse response begins at $n = 0$. Furthermore,

$$H(1) = \frac{5}{4}.$$

The poles of $H(z)$ are

$$p_k = \frac{1}{\sqrt{2}} \exp\left(j\frac{(2k-1)\pi}{4}\right), \quad \text{for } k = 1, 2, 3, 4.$$

The zeros of $H(z)$ are all at $z = 0$. Let

$$g[n] = j^n h[n].$$

The value of $g[8]$ equals _____. (Given the answer up to three decimal places.)

Q2.17.3 MCQ 2017 2M

Let a causal LTI system be characterized by the following differential equation, with initial rest condition

$$\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y(t) = 4x(t) + 5\frac{dx(t)}{dt}$$

where, $x(t)$ and $y(t)$ are the input and output respectively. The impulse response of the system is ($u(t)$ is the unit step function)

A. $2e^{-2t}u(t) - 7e^{-5t}u(t)$ **B.** $-2e^{-2t}u(t) + 7e^{-5t}u(t)$

C. $7e^{-2t}u(t) - 2e^{-5t}u(t)$ **D.** $-7e^{-2t}u(t) + 2e^{-5t}u(t)$

Q2.17.4 MCQ 2017 2M

A cascade system having the impulse responses $h_1(n) = \{1, -1\}$ and $h_2(n) = \{1, 1\}$ is shown in the figure below, where symbol $\uparrow$ denotes the time origin.

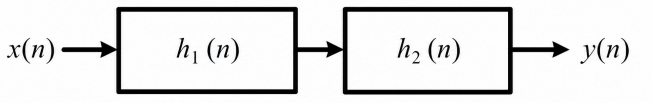


Fig. Q2.17.4

The input sequence $x(n)$ for which the cascade system produces an output sequence $y(n) = \{1, 2, 1, -1, -2, -1\}$ is

A. $x(n) = \{1, 2, 1, 1\}$ **B.** $x(n) = \{1, 1, 2, 2\}$

C. $x(n) = \{1, 1, 1, 1\}$ **D.** $x(n) = \{1, 2, 2, 1\}$

Q2.16.1 MCQ 2016 1M

Consider a continuous-time system with input $x(t)$ and output $y(t)$ given by

$$y(t) = x(t)\cos(t)$$

This system is

A. Linear and time-invariant

B. Non-linear and time-invariant

C. Linear and time-varying

D. Non-linear and time-varying

Q2.15.1 MCQ 2015 2M

For linear time invariant systems, that are Bounded Input Bounded Output stable, which one of the following statements is TRUE?

A. The impulse response will be integrable, but may not be absolutely integrable.

B. The unit impulse response will have finite support.

C. The unit step response will be absolutely integrable.

D. The unit step response will be bounded.

Q2.15.2 MCQ 2015 1M

The impulse response $g(t)$ of a system, G, is as shown in figure (a). What is the maximum value attained by the impulse response of two cascaded blocks of G as shown in figure (b)?

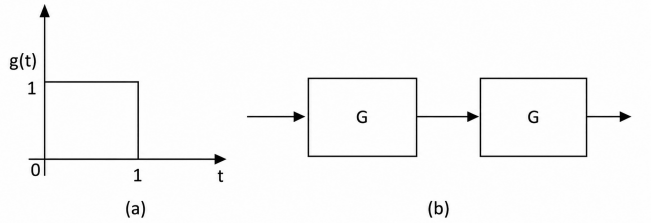


Fig. Q2.15.2

A. $\frac{2}{3}$ **B.** $\frac{3}{4}$ **C.** $\frac{4}{5}$ **D.** 1

Q2.14.1 MCQ 2014 1M

$x(t)$ is nonzero only for $T_x < t < T_x'$, and similarly, $y(t)$ is nonzero only for $T_y < t < T_y'$. Let $z(t)$ be convolution of $x(t)$ and $y(t)$. Which one of the following statements is TRUE?

A. $z(t)$ can be nonzero over an unbounded interval.

B. $z(t)$ is nonzero for $t < T_x + T_y$.

C. $z(t)$ is zero outside of $T_x + T_y < t < T_x' + T_y'$.

D. $z(t)$ is nonzero for $t < T_x' + T_y'$.

Q2.13.1 MCQ 2013 2M

The impulse response of a continuous time system is given by

$$h(t) = \delta(t-1) + \delta(t-3).$$

The value of the step response at $t = 2$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q2.13.2 MCQ 2013 1M

Two systems with impulse responses $h_1(t)$ and $h_2(t)$ are connected in cascade. Then the overall impulse response of the cascade system is given by

A. Product of $h_1(t)$ and $h_2(t)$

B. Sum of $h_1(t)$ and $h_2(t)$

C. Convolution of $h_1(t)$ and $h_2(t)$

D. Subtraction of $h_2(t)$ from $h_1(t)$

Q2.12.1 MCQ 2012 2M

The input $x(t)$ and output $y(t)$ of a system are related as

$$y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau.$$

The system is

A. Time-invariant and stable

B. Stable and not time-invariant

C. Time-invariant and not stable

D. Not time-invariant and not stable

Q2.12.2 MCQ 2012 2M

Let $y[n]$ denote the convolution of $h[n]$ and $g[n]$, where

$$h[n] = \left(\frac{1}{2}\right)^n u[n]$$

and $g[n]$ is a causal sequence. If $y[0] = 1$ and $y[1] = \frac{1}{2}$, then $g[1]$ equals

A. 0 **B.** $\frac{1}{2}$ **C.** 1 **D.** $\frac{3}{2}$

Q2.11.1 MCQ 2011 2M

The response $h(t)$ of a linear time invariant system to an impulse $\delta(t)$, under initially relaxed condition is

$$h(t) = \left(e^{-t} + e^{-2t}\right)u(t).$$

The response of this system for a unit step input $u(t)$ is

A. $u(t) + e^{-t} + e^{-2t}$ **B.** $\left(e^{-t} + e^{-2t}\right)u(t)$

C. $\left(1.5 - e^{-t} - 0.5e^{-2t}\right)u(t)$ **D.** $e^{-t}\delta(t) + e^{-2t}u(t)$

Q2.10.1 MCQ 2010 2M

Given the finite length input $x[n]$ and the corresponding finite length output $y[n]$ of an LTI system as shown below, the impulse response $h[n]$ of the system is

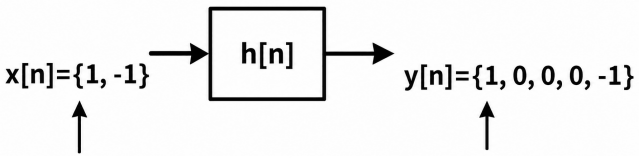


Fig. Q2.10.1

A. $h[n] = \left\{\underset{\uparrow}{1}\ 0\ 0\ 1\right\}$ **B.** $h[n] = \left\{\underset{\uparrow}{1}\ 0\ 1\right\}$

C. $h[n] = \left\{\underset{\uparrow}{1}\ 1\ 1\ 1\right\}$ **D.** $h[n] = \left\{\underset{\uparrow}{1}\ 1\ 1\right\}$

Q2.10.2 MCQ 2010 1M

The system represented by the input-output relationship

$$y(t) = \int_{-\infty}^{5t} x(\tau)\,d\tau, \qquad t > 0$$

A. Linear and causal **B.** Linear but not causal

C. Causal but not linear **D.** Neither linear nor causal

Q2.09.1 MCQ 2009 2M

A cascade of 3 Linear Time Invariant systems is causal and unstable. From this, we conclude that

A. Each system in the cascade is individually causal and unstable

B. At least one system is unstable and at least one system is causal

C. At least one system is causal and all systems are unstable

D. The majority are unstable and the majority are causal

Q2.09.2 MCQ 2009 1M

A Linear Time Invariant system with an impulse response $h(t)$ produces output $y(t)$ when the input $x(t)$ is applied. When the input $x(t-\tau)$ is applied to a system with impulse response $h(t-\tau)$, the output will be

A. $y(t)$ **B.** $y(2(t-\tau))$ **C.** $y(t-\tau)$ **D.** $y(t-2\tau)$

Q2.08.1 MCQ 2008 2M

A system with input $x(t)$ and output $y(t)$ is defined by the input-output relation: $y(t) = \int_{-\infty}^{-2t} x(\tau)\, d\tau$
The system will be

A. Causal, time-invariant and unstable

B. Causal, time-invariant and stable

C. Non-causal, time-invariant and unstable

D. Non-causal, time-variant and unstable

Q2.08.2 MCQ 2008 1M

The Impulse response of a causal linear time variant system is given as $h(t)$. Now consider the following two statements:
Statement (I): Principle of superposition holds
Statement (II): $h(t) = 0$ for $t < 0$.
Which one of the following statements is correct?

A. Statement (I) is correct and Statement (II) is wrong

B. Statement (II) is correct and Statement (I) is wrong

C. Both Statement (I) and Statement (II) are wrong

D. Both Statement (I) and Statement (II) are correct

Q2.08.3 MCQ 2008 1M

A signal $e^{-\alpha t}\sin(\omega t)$ is the input to a real Linear Time Invariant system. Given K and ϕ are constants, the output of the system will be of the form $Ke^{-\beta t}\sin(\nu t + \phi)$ where

A. β need not be equal to α but ν equal to ω

B. ν need not be equal to ω but β equal to α

C. β equal to α and ν equal to ω

D. β need not be equal to α and ν need not be equal to ω

Q2.07.1 MCQ 2007 1M

Let a signal $a_1\sin(\omega_1 t + \phi_1)$ be applied to a stable linear time-invariant system. Let the corresponding steady state output be represented as $a_2F(\omega_2 t + \phi_2)$. Then which of the following statements is true?

A. F is not necessarily a "sine" or "cosine" function but must be periodic with $\omega_1 = \omega_2$.

B. F must be a "sine" or "cosine" function with $a_1 = a_2$.

C. F must be a "sine" function with $\omega_1 = \omega_2$ and $\phi_1 = \phi_2$.

D. F must be a "sine" or "cosine" function with $\omega_1 = \omega_2$.

Q2.06.1 MCQ 2006 2M

$x[n] = 0;\ n < -1 \& n > 0,\ x[-1] = -1,\ x[0] = 2$ is the input and $y[n] = 0;\ n < -1 \& n > 2,\ y[-1] = -1 = y[1],\ y[0] = 3,\ y[2] = -2$ is the output of a discrete time LTI system. The system impulse response $h[n]$ will be

A. $h[n] = 0;\ n < 0,\ n > 2,\ h[0] = 1,\ h[1] = h[2] = -1$

B. $h[n] = 0;\ n < -1,\ n > 1,\ h[-1] = 1,\ h[0] = h[1] = 2$

C. $h[n] = 0;\ n < 0,\ n > 3,\ h[0] = -1,\ h[1] = 2,\ h[2] = 1$

D. $h[n] = 0;\ n < -2,\ n > 1,\ h[-2] = h[1] = h[-1] = -h[0] = 3$

Q2.06.2 MCQ 2006 2M

A continuous time system is described by $y(t) = e^{-|x(t)|}$, where $y(t)$ is the output and $x(t)$ is the input. $y(t)$ is bounded

A. only when $x(t)$ is bounded.

B. only when $x(t)$ is non-negative.

C. only for $t \geq 0$ if $x(t)$ is bounded for $t \geq 0$.

D. even when $x(t)$ is not bounded.

Q2.02.1 MCQ 2002 1M

Let $s(t)$ be the step response of a linear system with zero initial conditions; then the response of this system to an input $u(t)$ is

A. $\int_0^t s(t-\tau)\, u(\tau)\, d\tau$

B. $\frac{d}{dt}\left[\int_0^t s(t-\tau)\, u(\tau)\, d\tau\right]$

C. $\int_0^t s(t-\tau)\left[\int_0^\tau u(\tau_1)\, d\tau_1\right] d\tau$

D. $\int_0^t s(t-\tau)^2\, u(\tau)\, d\tau$

Q2.93.1 MCQ 1993 1M

$s(t)$ is step response and $h(t)$ is impulse response of a system. Its response $y(t)$ for any input $u(t)$ is given by

A. $\frac{d}{dt}\int_0^t s(t-\tau)u(\tau)\, d\tau$

B. $\int_0^t s(t-\tau)u(\tau)\, d\tau$

C. $\int_0^t \int_0^t s(t-\tau_1)u(\tau_1)\, d\tau_1\, d\tau$

D. $\frac{d}{dt}\int_0^t h(t-\tau)u(\tau)\, d\tau$

Q2.92.1 NAT 1992 5M

The impulse response of a network is $h(t) = 1$ for $0 \leq t < 1$ and zero otherwise. Sketch the impulse response of two such networks in cascade, neglecting loading effects.

Answer Key

Q2.25.1 B	Q2.25.2 A	Q2.25.3 B
Q2.25.4 A	Q2.24.1 D	Q2.24.2 B
Q2.24.3 A	Q2.23.1 B	Q2.23.2 4
Q2.22.1 B	Q2.22.2 D	Q2.21.1 C
Q2.21.2 D	Q2.20.1 C	Q2.20.2 C
Q2.20.3 A	Q2.19.1 D	Q2.18.1 A
Q2.17.1 D	Q2.17.2 0.09 to 0.1	Q2.17.3 B
Q2.17.4 D	Q2.16.1 C	Q2.15.1 D
Q2.15.2 D	Q2.14.1 C	Q2.13.1 B
Q2.13.2 C	Q2.12.1 D	Q2.12.2 A
Q2.11.1 C	Q2.10.1 C	Q2.10.2 B
Q2.09.1 B	Q2.09.2 D	Q2.08.1 D
Q2.08.2 D	Q2.08.3 C	Q2.07.1 D
Q2.06.1 A	Q2.06.2 D	Q2.02.1 B
Q2.93.1 A	Q2.92.1 *	

Note: * descriptive/evidence-based, no single definitive answer.

UNIT

III

Continuous Time Fourier Series

Topics

1. Introduction to Fourier Series
2. Exponential Fourier Series
3. Predicting the Nature of a Signal from Fourier Coefficients
4. Power Using Fourier Series Coefficients
5. Properties of Exponential Fourier Series
6. Fourier Series of Periodic Waveforms
7. Trigonometric Fourier Series and Symmetry
8. Polar Form, Dirichlet's Conditions and Gibbs Phenomenon

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 21	**MCQ:** 20	**MSQ:** 0	**NAT:** 1
1M: 9	**2M:** 10	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2017)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

A time-limited waveform $g(x)$ is specified as follows:

$$g(x) = \begin{cases} -k, & -\pi < x \le 0, \\ +k, & 0 < x \le \pi, \\ 0, & \text{otherwise.} \end{cases}$$

A new waveform $f(x)$ is constructed from $g(x)$ as follows:

$$f(x) = \sum_{m=-\infty}^{\infty} g(x + 2\pi m), \qquad \text{for all } x \in \mathbb{R}.$$

The sum of the coefficients of the third harmonics of the sine and cosine terms in the trigonometric Fourier series expansion of $f(x)$ is $\frac{2}{3\pi}$.
What is the value of k?

A. 1 **B.** $\frac{1}{2}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{4}$

Q3.19.1 MCQ 2019 2M

A periodic function $f(t)$, with a period of 2π, is represented as its Fourier series,

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt.$$

If

$$f(t) = \begin{cases} A \sin t, & 0 \le t \le \pi, \\ 0, & \pi < t < 2\pi, \end{cases}$$

The Fourier series coefficients a_1 and b_1 of $f(t)$ are

A. $a_1 = 0;\ b_1 = \frac{A}{2}$

B. $a_1 = \frac{A}{2};\ b_1 = 0$

C. $a_1 = \frac{A}{\pi};\ b_1 = 0$

D. $a_1 = 0;\ b_1 = \frac{A}{\pi}$

Q3.17.1 NAT 2017 1M

Consider

$$g(t) = \begin{cases} t - \lfloor t \rfloor, & t \ge 0, \\ t - \lceil t \rceil, & t < 0, \end{cases}$$

Here, $\lfloor t \rfloor$ represents the largest integer less than or equal to t and $\lceil t \rceil$ denotes the smallest integer greater than or equal to t. The coefficient of the second harmonic component of the Fourier series representing $g(t)$ is _____

Answer-checking note: This item is marked MTA (marks to all) in the supplied key. The stated two-sided function is not periodic, so the requested Fourier-series coefficient is not a valid numerical answer. A normal numeric check is inappropriate for the preserved question.

Q3.16.1 MCQ 2016 1M

Let $f(x)$ be a real, periodic function satisfying $f(-x) = -f(x)$. The general form of its Fourier series representation would be

A. $f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx)$

B. $f(x) = \sum_{k=1}^{\infty} b_k \sin(kx)$

C. $f(x) = a_0 + \sum_{k=1}^{\infty} a_{2k} \cos(kx)$

D. $f(x) = \sum_{k=0}^{\infty} a_{2k+1} \sin\big((2k+1)x\big)$

Q3.16.2 MCQ 2016 2M

The output of a continuous-time, linear time-invariant system is denoted by $T\{z(t)\}$ where $x(t)$ is the input signal. A signal $z(t)$ is called Eigen-signal of the system T, when

$$T\{z(t)\} = \gamma z(t),$$

where γ is a complex number, in general, and is called an Eigen value of T. Suppose the impulse response of the system T is real and even. Which of the following statements is TRUE?

A. $\cos(t)$ is an Eigen-signal but $\sin(t)$ is not

B. $\cos(t)$ and $\sin(t)$ are both Eigen-signals but with different Eigen values

C. $\sin(t)$ is an Eigen-signal but $\cos(t)$ is not

D. $\cos(t)$ and $\sin(t)$ are both Eigen-signals with identical Eigen values

Q3.15.1 MCQ 2015 2M

The signum function is given by

$$\text{sgn}(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

The Fourier series expansion of $\text{sgn}(\cos(t))$ has

A. only sine terms with all harmonics.

B. only cosine terms with all harmonics.

C. only sine terms with even numbered harmonics.

D. only cosine terms with odd numbered harmonics.

Q3.14.1 MCQ 2014 2M

A 10 kHz even-symmetric square wave is passed through a band-pass filter with center frequency at 30 kHz and 3 dB pass-band of 6 kHz. The filter output is

A. A highly attenuated square wave at 10 kHz

B. Nearly zero.

C. A nearly perfect cosine wave at 30 kHz

D. A nearly perfect sine wave at 30 kHz

Q3.14.2 MCQ 2014 1M

For a periodic square wave, which one of the following statements is TRUE?

A. The Fourier series coefficients do not exist.

B. The Fourier series coefficients exist but the reconstruction converges at no point.

C. The Fourier series coefficients exist and the reconstruction converges at most points.

D. The Fourier series coefficients exist and the reconstruction converges at every point.

Q3.13.1 MCQ 2013 1M

For a periodic signal

$$v(t) = 30\sin 100t + 10\cos 300t + 6\sin\left(500t + \frac{\pi}{4}\right),$$

the fundamental frequency in rad/s is

A. 100 **B.** 300 **C.** 500 **D.** 1500

Q3.11.1 MCQ 2011 1M

The Fourier series expansion

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + b_n \sin nt$$

of the periodic signal shown below will contain the following nonzero terms.

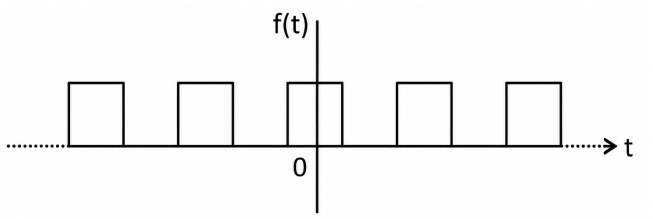


Fig. Q3.11.1

A. a_0 and b_n, $n = 1, 3, 5, \ldots, \infty$

B. a_0 and a_n, $n = 1, 2, 3, \ldots, \infty$

C. a_0, a_n and b_n, $n = 1, 2, 3, \ldots, \infty$

D. a_0 and a_n, $n = 1, 3, 5, \ldots, \infty$

Q3.10.1 MCQ 2010 1M

The second harmonic component of the periodic waveform given in the figure has amplitude of

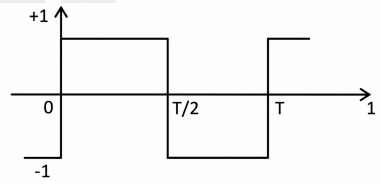


Fig. Q3.10.1

A. 0 **B.** 1 **C.** $\frac{2}{\pi}$ **D.** $\sqrt{5}$

Q3.09.1 MCQ 2009 2M

The Fourier Series coefficients, of a periodic signal $x(t)$, expressed as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{\frac{j2\pi kt}{T}}$$

are given by $a_{-2} = 2 - j$, $a_{-1} = 0.5 + j0.2$, $a_0 = j2$, $a_1 = 0.5 - j0.2$, $a_2 = 2 + j1$; and $a_k = 0$ for $|k| > 2$. Which of the following is true?

A. $x(t)$ has finite energy because only finitely many coefficients are non-zero

B. $x(t)$ has zero average value because it is periodic

C. The imaginary part of $x(t)$ is constant

D. The real part of $x(t)$ is even

Q3.08.1 MCQ 2008 2M

Let $x(t)$ be a periodic signal with time period T. Let

$$y(t) = x(t - t_0) + x(t + t_0)$$

for some t_0. The Fourier Series coefficients of $y(t)$ are denoted by b_k. If $b_k = 0$ for all odd k, then t_0 can be equal to

A. $\frac{T}{8}$ **B.** $\frac{T}{4}$ **C.** $\frac{T}{2}$ **D.** $2T$

Q3.07.1 MCQ 2007 2M

A signal $x(t)$ is given by

$$x(t) = \begin{cases} 1, & -\frac{T}{4} < t \le \frac{3T}{4}, \\ -1, & \frac{3T}{4} < t \le \frac{7T}{4}, \\ -x(t+T), & \text{otherwise} \end{cases}$$

Which among the following gives the fundamental Fourier term of $x(t)$?

A. $\frac{4}{\pi}\cos\left(\frac{\pi t}{T} - \frac{\pi}{4}\right)$ **B.** $\frac{4}{\pi}\cos\left(\frac{\pi t}{2T} + \frac{\pi}{4}\right)$

C. $\frac{4}{\pi}\sin\left(\frac{\pi t}{T} - \frac{\pi}{4}\right)$ **D.** $\frac{4}{\pi}\sin\left(\frac{\pi t}{2T} + \frac{\pi}{4}\right)$

Q3.06.1 MCQ 2006 1M

$x(t)$ is a real valued function of a real variable with period T. Its trigonometric Fourier Series expansion contains no terms of frequency $\omega = 2\pi(2k)/T$; $k = 1, 2, \ldots$. Also, no sine terms are present. Then $x(t)$ satisfies the equation

A. $x(t) = -x(t - T)$

B. $x(t) = x(T - t) = -x(-t)$

C. $x(t) = x(T - t) = -x\left(-t - \frac{T}{2}\right)$

D. $x(t) = x(t - T) = x\left(t - \frac{T}{2}\right)$

Q3.05.1 MCQ 2005 2M

The Fourier series for the function $f(x) = \sin^2 x$ is:

A. $\sin x + \sin 2x$ **B.** $1 - \cos 2x$

C. $\sin 2x + \cos 2x$ **D.** $0.5 - 0.5\cos 2x$

Q3.02.1 MCQ 2002 1M

Fourier Series for the waveform, $f(t)$ shown in Figure is

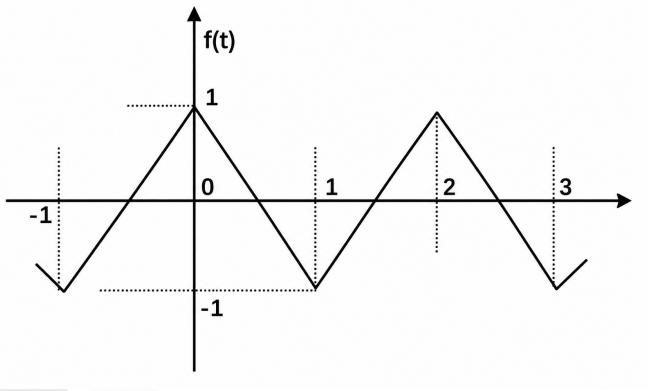


Fig. Q3.02.1

A. $\frac{8}{\pi^2}\left[\sin(\pi t) + \frac{1}{9}\sin(3\pi t) + \frac{1}{25}\sin(5\pi t) + \cdots\right]$

B. $\frac{8}{\pi^2}\left[\sin(\pi t) - \frac{1}{9}\sin(3\pi t) + \frac{1}{25}\sin(5\pi t) + \cdots\right]$

C. $\frac{8}{\pi^2}\left[\cos(\pi t) + \frac{1}{9}\cos(3\pi t) + \frac{1}{25}\cos(5\pi t) + \cdots\right]$

D. $\frac{8}{\pi^2}\left[\cos(\pi t) - \frac{1}{9}\sin(3\pi t) + \frac{1}{25}\sin(5\pi t) + \cdots\right]$

Q3.01.1 MCQ 2001 5M

Consider the voltage waveform v, shown in figure. Find

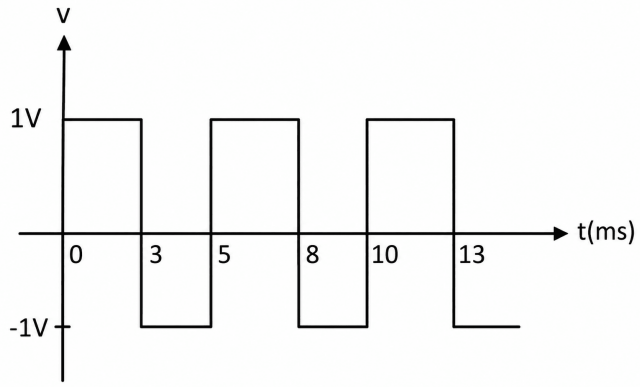


Fig. Q3.01.1

A. The dc component of v.

B. The amplitude of the fundamental component of v, and

C. The RMS value of the ac part of v.

Q3.00.1 MCQ 2000 2M

If an ac voltage wave is corrupted with an arbitrary number of harmonics, then the overall voltage waveform differs from its fundamental frequency component in terms of

A. Only the peak values

B. Only the RMS values

C. Only the average values

D. All the three measures (peak, RMS and average values)

Q3.97.1 MCQ 1997 5M

Compute the amplitude of the fundamental component of the waveform given in figure.

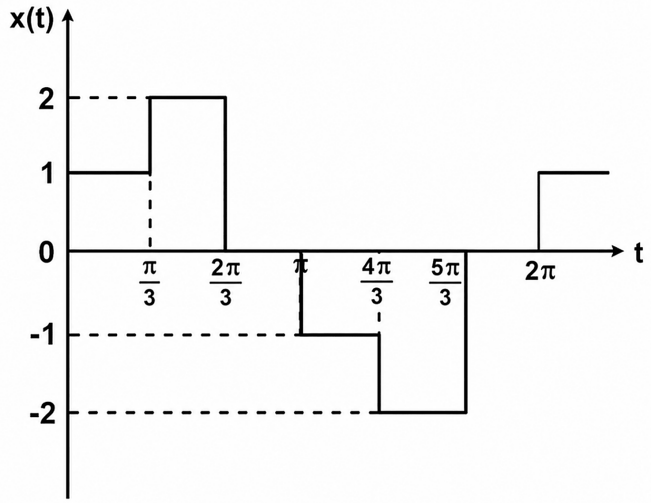


Fig. Q3.97.1

Q3.96.1 MCQ 1996 1M

A periodic rectangular signal, $x(t)$ has the wave form shown in figure. Frequency of the fifth harmonic of its spectrum is

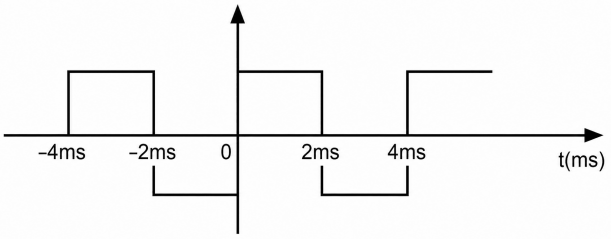


Fig. Q3.96.1

A. 40 Hz **B.** 200 Hz **C.** 250 Hz **D.** 1250 Hz

Answer Key

Note: **MTA** Marks To All awarded — see detailed solutions.

Q3.26.1 B	Q3.19.1 A	Q3.17.1 MTA
Q3.16.1 B	Q3.16.2 D	Q3.15.1 D
Q3.14.1 C	Q3.14.2 C	Q3.13.1 A
Q3.11.1 D	Q3.10.1 A	Q3.09.1 C
Q3.08.1 B	Q3.07.1 A	Q3.06.1 C
Q3.05.1 D	Q3.02.1 C	Q3.01.1 0.2, 1.21, 0.98
Q3.00.1 D	Q3.97.1 1.684	Q3.96.1 D

PrepFusion

UNIT

IV

Continuous Time Fourier Transform

Topics

1. Introduction to Fourier Transform
2. Properties of Fourier Transform
3. Fourier Transform of Standard Signals
4. Duality Property
5. FT of Rectangular, Triangular and Sampling Signals
6. Fourier Transform of Periodic Signals
7. Differentiation, Integration, Multiplication and Convolution Property
8. Parseval's Theorem
9. LTI Systems and Analog Filters
10. Nyquist Rate - Baseband and Bandpass Signals
11. Sampling Theorem and Applications

Unit Snapshot*

Stream: EE	**Questions:** 32	**MCQ:** 25	**MSQ:** 0	**NAT:** 7
1M: 12	**2M:** 20	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.25.1 NAT 2025 2M

An ideal low pass filter has frequency response given by

$$H(j\omega) = \begin{cases} 1, & |\omega| \le 200\pi, \\ 0, & \text{otherwise.} \end{cases}$$

Let $h(t)$ be its time domain representation. Then

$$h(0) = _____$$

(round off to the nearest integer).

Q4.24.1 NAT 2024 1M

Let $X(\omega)$ be the Fourier transform of the signal

$$x(t) = e^{-t^4}\cos t, \qquad -\infty < t < \infty.$$

The value of the derivative of $X(\omega)$ at $\omega = 0$ is _____ (rounded off to 1 decimal place).

Q4.23.1 MCQ 2023 1M

The Fourier transform $X(\omega)$ of the signal $x(t)$ is given by

$$X(\omega) = \begin{cases} 1, & \text{for } |\omega| < W_0, \\ 0, & \text{for } |\omega| > W_0. \end{cases}$$

Which one of the following statements is true?

A. $x(t)$ tends to be an impulse as $W_0 \to \infty$.

B. $x(0)$ decreases as W_0 increases.

C. At $t = \dfrac{\pi}{2W_0}$, $x(t) = -\dfrac{1}{\pi}$.

D. At $t = \dfrac{\pi}{2W_0}$, $x(t) = \dfrac{1}{\pi}$.

Q4.23.2 NAT 2023 2M

A signal $x(t) = 2\cos(180\pi t)\cos(60\pi t)$ is sampled at 200 Hz and then passed through an ideal low pass filter having cut-off frequency of 100 Hz. The maximum frequency present in the filtered signal, in Hz is _____. (Round off to the nearest integer.)

Q4.22.1 MCQ 2022 2M

Let an input

$$x(t) = 2\sin(10\pi t) + 5\cos(15\pi t) + 7\sin(42\pi t) + 4\cos(45\pi t)$$

is passed through an LTI system having an impulse response,

$$h(t) = 2\left(\frac{\sin 10\pi t}{\pi t}\right)\cos(40\pi t).$$

The output of the system is

A. $2\sin(10\pi t) + 5\cos(15\pi t)$ **B.** $5\cos(15\pi t) + 7\sin(42\pi t)$

C. $7\sin(42\pi t) + 4\cos(45\pi t)$ **D.** $2\sin(10\pi t) + 4\cos(45\pi t)$

Q4.21.1 NAT 2021 2M

Consider a continuous-time signal $x(t)$ defined by

$$x(t) = 0 \text{ for } |t| > 1, \quad \text{and} \quad x(t) = 1 - |t| \text{ for } |t| \le 1.$$

Let the Fourier transform of $x(t)$ be defined as

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt.$$

The maximum magnitude of $X(\omega)$ is _____.

Q4.21.2 MCQ 2021 2M

Let $f(t)$ be an even function, i.e., $f(-t) = f(t)$ for all t. Let the Fourier transform of $f(t)$ be defined as

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt.$$

Suppose

$$\frac{dF(\omega)}{d\omega} = -\omega F(\omega)$$

for all ω and $F(0) = 1$. Then

A. $f(0) < 1$ **B.** $f(0) > 1$

C. $f(0) = 1$ **D.** $f(0) = 0$

Q4.20.1 MCQ 2020 2M

The causal realization of a system transfer function $H(s)$ having poles at $(2, -1)$, $(-2, 1)$ and zeroes at $(2, 1)$, $(-2, -1)$ will be

A. Stable, real, All pass

B. Unstable, complex, All pass

C. Unstable, real, High pass

D. Stable, complex, Low pass

Q4.18.1 NAT 2018 2M

The Fourier transform of a continuous-time signal $x(t)$ is given by

$$X(\omega) = \frac{1}{(10 + j\omega)^2}, \qquad -\infty < \omega < \infty,$$

where $j = \sqrt{-1}$ and ω denotes frequency. Then the value of $|\ln x(t)|$ at t=1 is _____ (up to 1 decimal place). (ln denotes the logarithm to base e)

Q4.17.1 MCQ 2017 2M

Let the signal

$$x(t) = \sum_{k=-\infty}^{+\infty} (-1)^k \delta\left(t - \frac{k}{2000}\right)$$

be passed through an LTI system with frequency response $H(\omega)$, as given in the figure below.

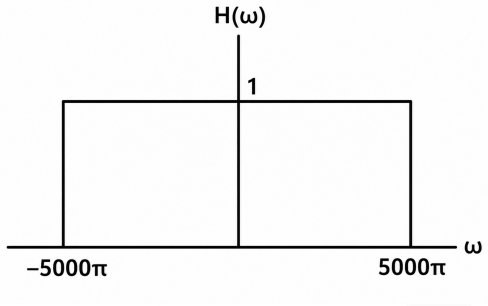


Fig. Q4.17.1

The Fourier series representation of the output signal is given as

A. $4000 + 4000\cos(2000\pi t) + 4000\cos(4000\pi t)$

B. $2000 + 2000\cos(2000\pi t) + 2000\cos(4000\pi t)$

C. $4000\cos(2000\pi t)$

D. $2000\cos(2000\pi t)$

Q4.17.2 MCQ 2017 1M

Let $z(t) = x(t) * y(t)$, where "$*$" denotes convolution. Let c be a positive real-valued constant. Choose the correct expression for $z(ct)$.

A. $c \cdot x(ct) * y(ct)$ **B.** $x(ct) * y(ct)$
C. $c \cdot x(t) * y(ct)$ **D.** $c \cdot x(ct) * y(t)$

Q4.17.3 MCQ 2017 2M

The output $y(t)$ of the following system is to be sampled, so as to reconstruct it from its samples uniquely. The required minimum sampling rate is

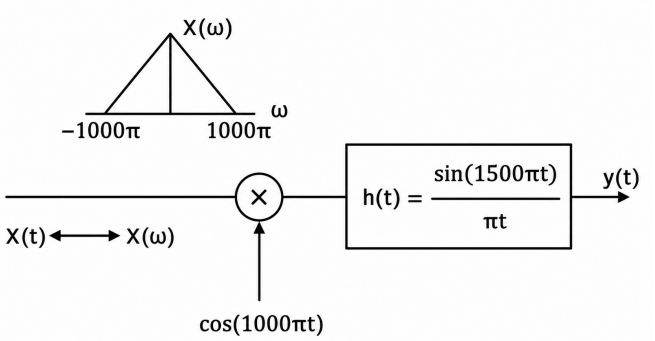


Fig. Q4.17.3

A. 1000 samples/s **B.** 1500 samples/s
C. 2000 samples/s **D.** 3000 samples/s

Q4.16.1 MCQ 2016 2M

The value of the integral

$$2\int_{-\infty}^{\infty} \left(\frac{\sin 2\pi t}{\pi t}\right) dt$$

is equal to

A. 0 **B.** 0.5 **C.** 1 **D.** 2

Q4.16.2 NAT 2016 1M

Suppose the maximum frequency in a band-limited signal $x(t)$ is 5 kHz. Then, the maximum frequency in $x(t)\cos(2000\pi t)$, in kHz, is _____.

Q4.16.3 MCQ 2016 2M

Suppose $x_1(t)$ and $x_2(t)$ have the Fourier transforms as shown below.

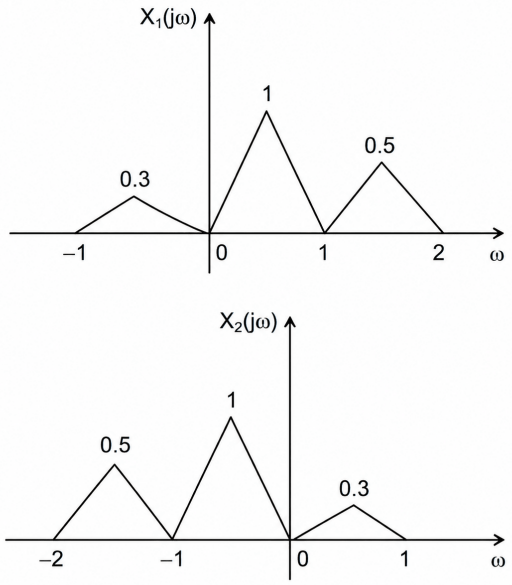


Fig. Q4.16.3

Which one of the following statements is TRUE?

A. $x_1(t)$ and $x_2(t)$ are complex and $x_1(t)x_2(t)$ is also complex with non-zero imaginary part

B. $x_1(t)$ and $x_2(t)$ are real and $x_1(t)x_2(t)$ is also real

C. $x_1(t)$ and $x_2(t)$ are complex but $x_1(t)x_2(t)$ is real

D. $x_1(t)$ and $x_2(t)$ are imaginary but $x_1(t)x_2(t)$ is real

Q4.16.4 MCQ 2016 2M

Let $x_1(t) \leftrightarrow X_1(\omega)$ and $x_2(t) \leftrightarrow X_2(\omega)$ be two signals whose Fourier Transforms are as shown in the figure shown below. In the figure, $h(t) = e^{-2|t|}$ denotes the impulse response.

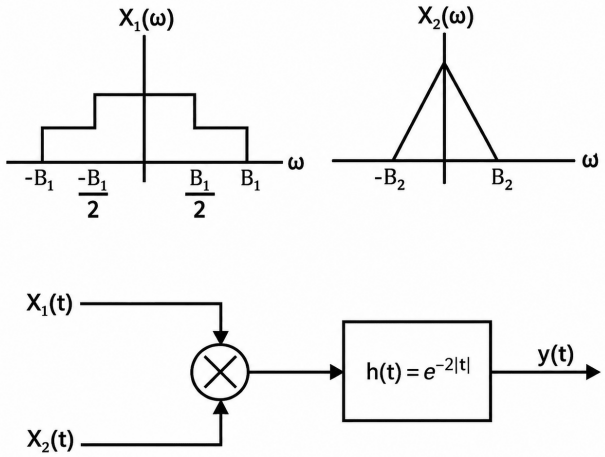


Fig. Q4.16.4

For the system shown above, the minimum sampling rate required to sample $y(t)$, so that $y(t)$ can be uniquely reconstructed from its samples, is

A. $2B_1$ **B.** $2(B_1 + B_2)$
C. $4(B_1 + B_2)$ **D.** ∞

Q4.15.1 MCQ 2015 1M

The filters F1 and F2 having characteristics as shown in figures (a) and (b) are connected as shown in figure (c).

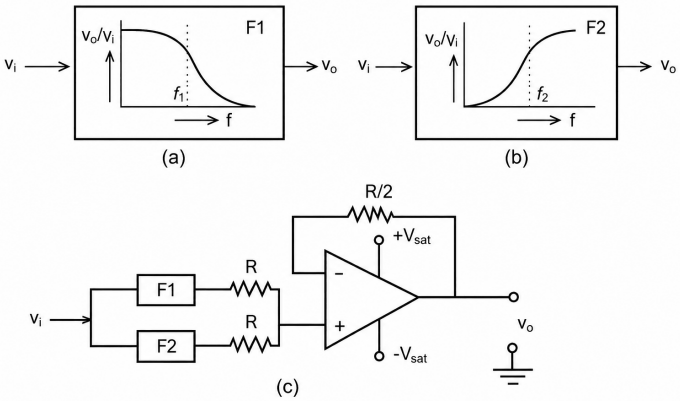


Fig. Q4.15.1

The cut-off frequencies of F1 and F2 are f_1 and f_2 respectively. If $f_1 < f_2$, the resultant circuit exhibit the characteristics of a

A. Band-pass filter **B.** Band-stop filter
C. All pass filter **D.** High-Q filter

Q4.15.2 MCQ 2015 2M

Consider a signal defined by

$$x(t) = \begin{cases} e^{j10t}, & \text{for } |t| \le 1, \\ 0, & \text{for } |t| > 1. \end{cases}$$

Its Fourier Transform is

A. $\dfrac{2\sin(\omega - 10)}{\omega - 10}$ **B.** $2e^{j10}\dfrac{\sin(\omega - 10)}{\omega - 10}$

C. $\dfrac{2\sin\omega}{\omega - 10}$ **D.** $e^{j10\omega}\dfrac{2\sin\omega}{\omega}$

Q4.14.1 MCQ 2014 2M

A differentiable non constant even function $x(t)$ has a derivative $y(t)$, and their respective Fourier Transformers are $X(\omega)$ and $Y(\omega)$. Which of the following statements is TRUE?

A. $X(\omega)$ and $Y(\omega)$ are both real.

B. $X(\omega)$ is real and $Y(\omega)$ is imaginary.

C. $X(\omega)$ and $Y(\omega)$ are both imaginary.

D. $X(\omega)$ is imaginary and $Y(\omega)$ is real.

Q4.14.2 MCQ 2014 2M

A continuous-time LTI system with system function $H(\omega)$ has the following pole-zero plot. For this system, which of the alternatives is TRUE?

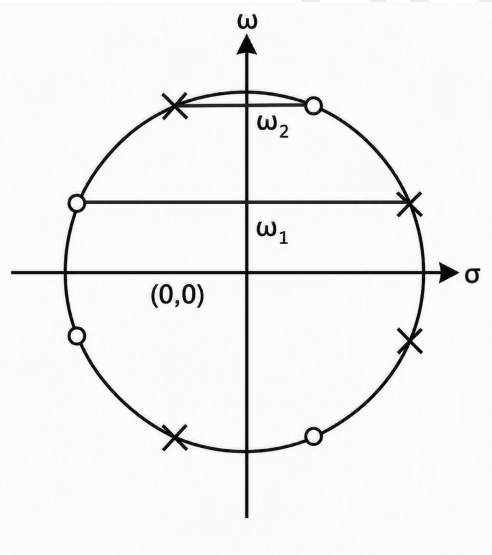


Fig. Q4.14.2

A. $|H(0)| > |H(\omega)|;\ |\omega| > 0$

B. $|H(\omega)|$ has multiple maxima, at ω_1 and ω_2

C. $|H(0)| < |H(\omega)|;\ |\omega| > 0$

D. $|H(\omega)|$ = constant; $-\infty < \omega < \infty$

Q4.14.3 MCQ 2014 1M

A signal is represented by

$$x(t) = \begin{cases} 1, & |t| < 1, \\ 0, & |t| > 1. \end{cases}$$

The Fourier transform of the convolved signal

$$y(t) = x(2t) * x(t/2)$$

is

A. $\dfrac{4}{\omega^2}\sin\left(\dfrac{\omega}{2}\right)\sin(2\omega)$ **B.** $\dfrac{4}{\omega^2}\sin\left(\dfrac{\omega}{2}\right)$

C. $\dfrac{4}{\omega^2}\sin(2\omega)$ **D.** $\dfrac{4}{\omega^2}\sin^2\omega$

Q4.14.4 MCQ 2014 1M

A function $f(t)$ is shown in the figure.

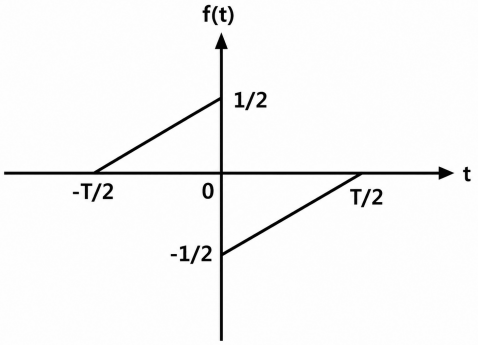


Fig. Q4.14.4

The Fourier transform $F(\omega)$ of $f(t)$ is

A. Real and even function of ω

B. Real and odd function of ω

C. Imaginary and odd function of ω

D. Imaginary and even function of ω

Q4.14.5 MCQ 2014 1M

Consider an LTI system with transfer function

$$H(s) = \frac{1}{s(s+4)}.$$

If the input to the system is $\cos(3t)$ and the steady state output is $A\sin(3t+\alpha)$, then the value of A is

A. $\frac{1}{30}$ **B.** $\frac{1}{15}$ **C.** $\frac{3}{4}$ **D.** $\frac{4}{3}$

Q4.14.6 MCQ 2014 2M

Let $f(t)$ be a continuous time signal and let $F(\omega)$ be its Fourier transform defined by

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt.$$

Define $g(t)$ by

$$g(t) = \int_{-\infty}^{\infty} F(\omega)e^{-j\omega t}\,d\omega.$$

What is the relationship between $f(t)$ and $g(t)$?

A. $g(t)$ would always be proportional to $f(t)$.

B. $g(t)$ would be proportional to $f(t)$ if $f(t)$ is an even function.

C. $g(t)$ would be proportional to $f(t)$ only if $f(t)$ is a sinusoidal function.

D. $g(t)$ would never be proportional to $f(t)$.

Q4.14.7 MCQ 2014 2M

A sinusoid $x(t)$ of unknown frequency is sampled by an impulse train of period 20 ms. The resulting sample train is next applied to an ideal low pass filter with a cut off at 25 Hz. The filter output is seen to be a sinusoid of frequency 20 Hz. This means that $x(t)$ has a frequency of

A. 10 Hz **B.** 60 Hz
C. 30 Hz **D.** 90 Hz

Q4.14.8 NAT 2014 1M

For the signal $f(t) = 3\sin 8\pi t + 6\sin 12\pi t + \sin 14\pi t$, the minimum sampling frequency (in Hz) satisfying the Nyquist criterion is _____.

Q4.13.1 MCQ 2013 1M

A band-limited signal with a maximum frequency of 5 kHz is to be sampled. According to the sampling theorem, the sampling frequency in kHz which is not valid is

A. 5 **B.** 12
C. 15 **D.** 20

Q4.12.1 MCQ 2012 2M

The Fourier transform of a signal $h(t)$ is

$$H(j\omega) = \frac{(2\cos\omega)(\sin 2\omega)}{\omega}.$$

The value of $h(0)$ is

A. $\frac{1}{4}$ **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Q4.10.1 MCQ 2010 2M

$x(t)$ is a positive rectangular pulse from $t=-1$ to $t=+1$ with unit height as shown in the figure. The value of

$$\int_{-\infty}^{\infty} |X(\omega)|^2\,d\omega$$

where $X(\omega)$ is the Fourier transform of $x(t)$ is

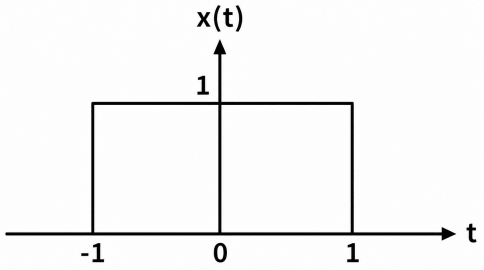


Fig. Q4.10.1

A. 2 **B.** 2π **C.** 4 **D.** 4π

Q4.08.1 MCQ 2008 2M

A signal

$$x(t) = \text{sinc}(\alpha t)$$

where α is a real constant

$$\left(\text{sinc}(x) = \frac{\sin \pi x}{\pi x}\right)$$

is the input to a Linear Time invariant system whose impulse response

$$h(t) = \text{sinc}(\beta t)$$

where β is a real constant. If $\min(\alpha, \beta)$ denotes the minimum of α and β, and similarly $\max(\alpha, \beta)$ denotes the maximum of α and β, and K is a constant, which one of the following statements is true about the output of the system?

A. It will be of the form $K\,\text{sinc}(\gamma t)$ where $\gamma = \min(\alpha, \beta)$

B. It will be of the form $K\,\text{sinc}(\gamma t)$ where $\gamma = \max(\alpha, \beta)$

C. It will be of the form $K\,\text{sinc}(\alpha t)$

D. It cannot be a sinc type of signal

Q4.07.1 MCQ 2007 1M

Let a signal $a_1 \sin(\omega_1 t + \phi_1)$ be applied to a stable linear time-invariant system. Let the corresponding steady state output be represented as $a_2 F(\omega_2 t + \phi_2)$. Then which of the following statements is true?

A. F is not necessarily a "sine" or "cosine" function but must be periodic with $\omega_1 = \omega_2$

B. F must be a "sine" or "cosine" function with $a_1 = a_2$

C. F must be a "sine" function with $\omega_1 = \omega_2$ and $\phi_1 = \phi_2$

D. F must be a "sine" or "cosine" function with $\omega_1 = \omega_2$

Q4.07.2 MCQ 2007 1M

The frequency spectrum of a signal is shown in the figure. If this signal is ideally sampled at intervals of 1 ms, then the frequency spectrum of the sampled signal will be

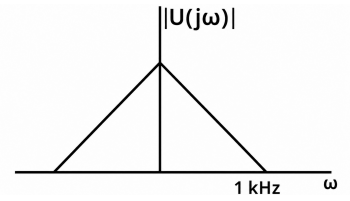
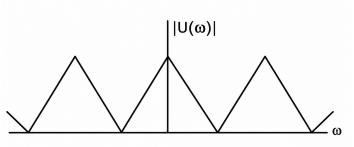
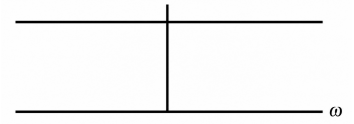


Fig. Q4.07.2

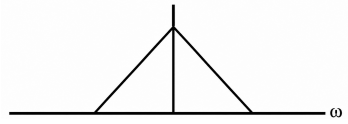
A.



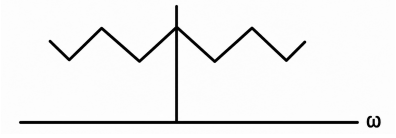
B.



C.



D.



Answer Key

Q4.25.1 200	Q4.24.1 0	Q4.23.1 A
Q4.23.2 80	Q4.22.1 C	Q4.21.1 1
Q4.21.2 A	Q4.20.1 b	Q4.18.1 10
Q4.17.1 C	Q4.17.2 A	Q4.17.3 B
Q4.16.1 D	Q4.16.2 6	Q4.16.3 C
Q4.16.4 B	Q4.15.1 B	Q4.15.2 A
Q4.14.1 B	Q4.14.2 D	Q4.14.3 A
Q4.14.4 C	Q4.14.5 B	Q4.14.6 B
Q4.14.7 C	Q4.14.8 14	Q4.13.1 A
Q4.12.1 C	Q4.10.1 D	Q4.08.1 A
Q4.07.1 D	Q4.07.2 B	

UNIT

V

Laplace Transform

Topics

1. Introduction to Laplace Transform
2. Laplace Transform of Standard Signals and Properties
3. Region of Convergence (ROC)
4. Inverse Laplace Transform
5. Interrelation Between EFSC, FT and LT
6. Eigen Values and Eigen Functions in LTI Systems
7. Unilateral Laplace Transform

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 22	**MCQ:** 21	**MSQ:** 0	**NAT:** 1
1M: 15	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2024)–4M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q5.24.1 MCQ 2024 1M

If $u(t)$ is the unit step function, then the region of convergence (ROC) of the Laplace transform of the signal

$$x(t) = e^{t^2}\,[u(t-1) - u(t-10)]$$

is

A. $-\infty < Re(s) < \infty$ **B.** $Re(s) \geq 10$

C. $Re(s) \leq 1$ **D.** $1 \leq Re(s) \leq 10$

Q5.20.1 MCQ 2020 1M

Which of the following statements is true about the two sided Laplace transform?

A. It exists for every signal that may or may not have a Fourier transform.

B. It has no poles for any bounded signal that is non-zero only inside a finite time interval.

C. The number of finite poles and finite zeroes must be equal.

D. If a signal can be expressed as a weighted sum of shifted one sided exponentials, then its Laplace Transform will have no poles.

Q5.19.1 MCQ 2019 1M

The inverse Laplace transform of

$$H(s) = \frac{s+3}{s^2+2s+1}$$

for $t \geq 0$ is

A. $3te^{-t} + e^{-t}$ **B.** $4te^{-t} + e^{-t}$

C. $3te^{-t}$ **D.** $2te^{-t} + e^{-t}$

Q5.19.2 MCQ 2019 1M

A system transfer function is

$$H(s) = \frac{a_1 s^2 + b_1 s + c_1}{a_2 s^2 + b_2 s + c_2}.$$

If $a_1 = b_1 = 0$, and all other coefficients are positive, the transfer function represents a

A. Band pass filter **B.** High pass filter

C. Notch filter **D.** Low pass filter

Q5.16.1 MCQ 2016 1M

Consider a causal LTI system characterized by differential equation

$$\frac{dy(t)}{dt} + \frac{1}{6}y(t) = 3x(t).$$

The response of the system to the input

$$x(t) = 3e^{-\frac{t}{3}}u(t),$$

where $u(t)$ denotes the unit step function, is

A. $9e^{-t/3}u(t)$ **B.** $9e^{-t/6}u(t)$

C. $9e^{-t/3}u(t) - 6e^{-t/6}u(t)$ **D.** $54e^{-t/6}u(t) - 54e^{-t/3}u(t)$

Q5.16.2 MCQ 2016 1M

The Laplace Transform of

$$f(t) = e^{2t}\sin(5t)u(t)$$

is

A. $\dfrac{5}{s^2-4s+29}$ **B.** $\dfrac{5}{s^2+5}$

C. $\dfrac{s-2}{s^2-4s+29}$ **D.** $\dfrac{5}{s+5}$

Q5.15.1 MCQ 2015 1M

The Laplace transform of

$$f(t) = 2\sqrt{\frac{t}{\pi}}$$

is $s^{-3/2}$. The Laplace transform of

$$g(t) = \sqrt{\frac{1}{\pi t}}$$

is

A. $\dfrac{3}{2}s^{-5/2}$ **B.** $s^{-1/2}$ **C.** $s^{1/2}$ **D.** $s^{3/2}$

Q5.14.1 MCQ 2014 1M

Consider an LTI system with impulse response

$$h(t) = e^{-5t}u(t).$$

If the output of the system is

$$y(t) = e^{-3t}u(t) - e^{-5t}u(t)$$

then the input, $x(t)$, is given by

A. $e^{-3t}u(t)$ **B.** $2e^{-3t}u(t)$

C. $e^{-5t}u(t)$ **D.** $2e^{-5t}u(t)$

Q5.14.2 MCQ 2014 1M

Let

$$X(s) = \frac{3s+5}{s^2+10s+21}$$

be the Laplace transform of a signal $x(t)$. Then, $x(0^+)$ is

A. 0 **B.** 3 **C.** 5 **D.** 21

Q5.13.1 MCQ 2013 1M

Which one of the following statements is NOT TRUE for a continuous time causal and stable LTI system?

A. All the poles of the system must lie on the left side of the $j\omega$ axis.

B. Zeros of the system can lie anywhere in the s-plane.

C. All the poles must lie within $|s| = 1$.

D. All the roots of the characteristic equation must be located on the left side of the $j\omega$ axis.

Q5.13.2 MCQ 2013 1M

The impulse response of a system is

$$h(t) = tu(t).$$

For an input $u(t-1)$, the output is

A. $\frac{t^2}{2}u(t)$ **B.** $\frac{t(t-1)}{2}u(t-1)$

C. $\frac{(t-1)^2}{2}u(t-1)$ **D.** $\frac{t^2-1}{2}u(t-1)$

Q5.12.1 MCQ 2012 1M

The unilateral Laplace transform of $f(t)$ is

$$\frac{1}{s^2+s+1}.$$

The unilateral Laplace transform of $tf(t)$ is

A. $-\frac{s}{(s^2+s+1)^2}$ **B.** $-\frac{2s+1}{(s^2+s+1)^2}$

C. $\frac{s}{(s^2+s+1)^2}$ **D.** $\frac{2s+1}{(s^2+s+1)^2}$

Q5.11.1 MCQ 2011 2M

Let the Laplace transform of a function $f(t)$ which exists for $t > 0$ be $F_1(s)$ and the Laplace transform of its delayed version $f(t-\tau)$ be $F_2(s)$. Let $F_1^*(s)$ be the complex conjugate of $F_1(s)$ with the Laplace variable set as $s = \sigma + j\omega$. If

$$G(s) = \frac{F_2(s) \cdot F_1^*(s)}{|F_1(s)|^2},$$

then the inverse Laplace transform of $G(s)$ is

A. An ideal impulse $\delta(t)$

B. An ideal delayed impulse $\delta(t-\tau)$

C. An ideal step function $u(t)$

D. An ideal delayed step function $u(t-\tau)$

Q5.11.2 MCQ 2011 1M

Given two continuous time signals

$$x(t) = e^{-t}$$

and

$$y(t) = e^{-2t}$$

which exist for $t > 0$, the convolution

$$z(t) = x(t) * y(t)$$

is

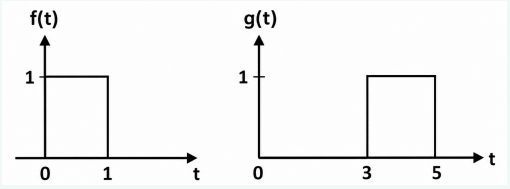
A. $e^{-t} - e^{-2t}$ **B.** e^{-3t}

C. e^{+t} **D.** $e^{-t} + e^{-2t}$

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Questions Q5.10.1 and Q5.10.2 Given $f(t)$ and $g(t)$ as shown below:



Q5.10.1 MCQ 2010 2M

$g(t)$ can be expressed as

A. $g(t) = f(2t-3)$ **B.** $g(t) = f\left(\frac{t}{2}-3\right)$

C. $g(t) = f\left(2t-\frac{3}{2}\right)$ **D.** $g(t) = f\left(\frac{t}{2}-\frac{3}{2}\right)$

Q5.10.2 MCQ 2010 2M

The Laplace transform of $g(t)$ is

A. $\frac{1}{s}\left(e^{3s}-e^{5s}\right)$ **B.** $\frac{1}{s}\left(e^{-5s}-e^{3s}\right)$

C. $\frac{e^{-3s}}{s}\left(1-e^{-2s}\right)$ **D.** $\frac{1}{s}\left(e^{2s}-e^{3s}\right)$

Q5.07.1 MCQ 2007 2M

If $u(t)$, $r(t)$ denote the unit step and unit ramp functions respectively and $u(t) * r(t)$ their convolution. Then the function

$$u(t+1) * r(t-2)$$

is given by

A. $\frac{1}{2}(t-1)(t-2)$ **B.** $\frac{1}{2}(t-1)(t-2)u(t-1)$

C. $\frac{1}{2}(t-1)^2u(t-1)$ **D.** None of the above

Q5.06.1 MCQ 2006 2M

The running integrator, given by

$$y(t)=\int_{-\infty}^{t} x(t')\,dt'$$

A. Has no finite singularities in its double sided Laplace Transform $Y(s)$.

B. Produces a bounded output for every causal bounded input.

C. Produces a bounded output for every anti causal bounded input.

D. Has no finite zeros in its double sided Laplace Transform $Y(s)$.

Q5.05.1 MCQ 2005 2M

The Laplace transform of a function $f(t)$ is

$$F(s)=\frac{5s^2+23s+6}{s(s^2+2s+2)}.$$

As $t \to \infty$, $f(t)$ approaches

A. 3 **B.** 5 **C.** $\frac{17}{2}$ **D.** ∞

Q5.02.1 MCQ 2002 1M

Let $Y(s)$ be the Laplace transformation of the function $y(t)$, then the final value of the function is

A. $\lim_{s\to 0} Y(s)$ **B.** $\lim_{s\to \infty} Y(s)$

C. $\lim_{s\to 0} sY(s)$ **D.** $\lim_{s\to \infty} sY(s)$

Q5.99.1 MCQ 1999 2M

A rectangular current pulse of duration T and magnitude I has the Laplace transform

A. $\frac{I}{s}$ **B.** $\left(\frac{I}{s}\right)\exp(-Ts)$

C. $\left(\frac{I}{s}\right)\exp(Ts)$ **D.** $\left(\frac{I}{s}\right)[1-\exp(-Ts)]$

Q5.95.1 NAT 1995 1M

The convolution of the functions

$$f_1(t)=e^{-2t}u(t)$$

and

$$f_2(t)=e^{-t}u(t)$$

is equal to _____.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q5.24.1 A	Q5.20.1 B	Q5.19.1 D
Q5.19.2 D	Q5.16.1 D	Q5.16.2 A
Q5.15.1 B	Q5.14.1 B	Q5.14.2 B
Q5.13.1 C	Q5.13.2 C	Q5.12.1 D
Q5.11.1 B	Q5.11.2 A	Q5.10.1 D
Q5.10.2 C	Q5.07.1 C	Q5.06.1 D
Q5.05.1 A	Q5.02.1 C	Q5.99.1 D
Q5.95.1 *		

UNIT

VI

Z Transform, DTFS and DTFT

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform
6. DTFT
7. Discrete-Time Fourier Series (DTFS)

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 24	**MCQ:** 23	**MSQ:** 0	**NAT:** 1
1M: 5	**2M:** 19	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 MCQ 2026 1M

Consider the infinite-length, discrete-time sequence

$$x[n] = 0.9^{|n|},$$

where n is an integer. The region of convergence of its Z-transform $X(z)$ is given by:
(Note: z is a complex variable)

A. $|z| > 0.9$

B. $|z| < 0.9$

C. $0.9 < |z| < \dfrac{1}{0.9}$

D. $\{z \text{ such that } |z| < 0.9\} \cup \{z \text{ such that } |z| > \dfrac{1}{0.9}\}$

Q6.24.1 MCQ 2024 2M

If the Z-transform of a finite-duration discrete-time signal $x[n]$ is $X(z)$, then the Z transform of the signal

$$y[n] = x[2n]$$

is

A. $Y(z) = X\left(z^2\right)$

B. $Y(z) = \dfrac{1}{2}\left[X\left(z^{-1/2}\right) + X\left(-z^{-1/2}\right)\right]$

C. $Y(z) = \dfrac{1}{2}\left[X\left(z^{1/2}\right) + X\left(-z^{1/2}\right)\right]$

D. $Y(z) = \dfrac{1}{2}\left[X\left(z^2\right) + X\left(-z^2\right)\right]$

Q6.23.1 MCQ 2023 1M

The Z-transform of a discrete signal $x[n]$ is

$$X(z) = \frac{4z}{\left(z - \frac{1}{5}\right)\left(z - \frac{2}{3}\right)(z - 3)}$$

with ROC = R.
Which one of the following statements is true?

A. Discrete-time Fourier transform of $x[n]$ converges if R is $|z| > 3$.

B. Discrete-time Fourier transform of $x[n]$ converges if R is $\dfrac{2}{3} < |z| < 3$.

C. Discrete-time Fourier transform of $x[n]$ converges if R is such that $x[n]$ is a left-sided sequence.

D. Discrete-time Fourier transform of $x[n]$ converges if R is such that $x[n]$ is a right-sided sequence.

Q6.23.2 NAT 2023 2M

The discrete-time Fourier transform of a signal $x[n]$ is

$$X(\Omega) = (1 + \cos\Omega)e^{-j\Omega}.$$

Consider that $x_p[n]$ is a periodic signal of period $N = 5$ such that

$$x_p[n] = \begin{cases} x[n], & \text{for } n = 0, 1, 2, \\ 0, & \text{for } n = 3, 4. \end{cases}$$

Note that

$$x_p[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}kn}.$$

The magnitude of the Fourier series coefficient a_3 is _____ (Round off to 3 decimal places).

Q6.22.1 MCQ 2022 2M

The discrete time Fourier series representation of a signal $x[n]$ with period N is written as

$$x[n] = \sum_{k=0}^{N-1} a_k e^{j\left(\frac{2\pi kn}{N}\right)}.$$

A discrete time periodic signal with period $N = 3$, has the non-zero Fourier series coefficients: $a_{-3} = 2$ and $a_4 = 1$. The signal is

A. $2 + 2e^{-j\left(\frac{2\pi}{6}n\right)}\cos\left(\frac{2\pi}{6}n\right)$

B. $1 + 2e^{j\left(\frac{2\pi}{6}n\right)}\cos\left(\frac{2\pi}{6}n\right)$

C. $1 + 2e^{j\left(\frac{2\pi}{3}n\right)}\cos\left(\frac{2\pi}{6}n\right)$

D. $2 + 2e^{j\left(\frac{2\pi}{6}n\right)}\cos\left(\frac{2\pi}{6}n\right)$

Q6.21.1 MCQ 2021 2M

The causal signal with z-transform

$$\frac{z^2}{(z-a)^2}$$

is ([$u[n]$ is the unit step signal])

A. $a^2 n\, u[n]$ **B.** $(n+1)a^n u[n]$

C. $n^{-1}a^n u[n]$ **D.** $n^2 a^n u[n]$

Q6.20.1 MCQ 2020 1M

Consider a signal

$$x[n] = \left(\frac{1}{2}\right)^n u[n],$$

where $u[n] = 0$ if $n < 0$, and $u[n] = 1$ if $n \geq 0$. The z-transform of $x[n-k]$, $k > 0$ is

$$\frac{z^{-k}}{1 - \frac{1}{2}z^{-1}}$$

with region of convergence being

A. $|z| < 2$ **B.** $|z| > 2$ **C.** $|z| < \frac{1}{2}$ **D.** $|z| > \frac{1}{2}$

Q6.17.1 MCQ 2017 1M

The pole-zero plots of three discrete-time systems P, Q and R on the z-plane are shown below.

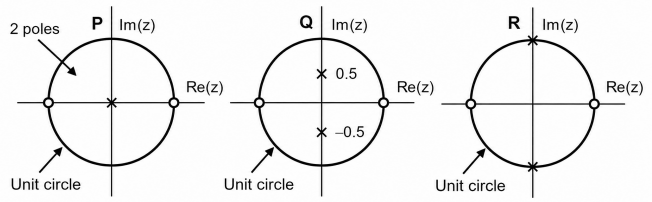


Fig. Q6.17.1

Which one of the following is TRUE about the frequency selectivity of these systems?

A. All three are high-pass filters.

B. All three are band-pass filters.

C. All three are low-pass filters.

D. P is a low-pass filter, Q is a band-pass filter and R is a high-pass filter.

Q6.17.2 MCQ 2017 2M

A cascade system having the impulse responses $h_1(n) = \{\underset{\uparrow}{1}, -1\}$ and $h_2(n) = \{\underset{\uparrow}{1}, 1\}$ is shown in the figure below, where symbol $\uparrow$ denotes the time origin.

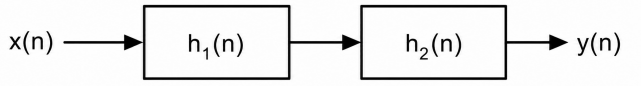


Fig. Q6.17.2

The input sequences $x(n)$ for which the cascade system produces an output sequence $y(n) = \{1, 2, 1, -1, -2, -1\}$ is

A. $x(n) = \{\underset{\uparrow}{1}, 2, 1, 1\}$ **B.** $x(n) = \{\underset{\uparrow}{1}, 1, 2, 2\}$

C. $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1\}$ **D.** $x(n) = \{\underset{\uparrow}{1}, 2, 2, 1\}$

Q6.15.1 MCQ 2015 2M

Consider a discrete time signal given by

$$x[n] = (-0.25)^n u[n] + (0.5)^n u[-n-1].$$

The region of convergence of its Z-transform would be

A. The region inside the circle of radius 0.5 and centered at origin.

B. The region outside the circle of radius 0.25 and centered at origin.

C. The annular region between the two circles, both centered at origin and having radii 0.25 and 0.5.

D. The entire Z plane.

Q6.15.2 MCQ 2015 2M

The z-transform of a sequence $x[n]$ is given as

$$X(z) = 2z + 4 - \frac{4}{z} + \frac{3}{z^2}.$$

If $y[n]$ is the first difference of $x[n]$, then $Y(z)$ is given by

A. $2z + 2 - \frac{8}{z} + \frac{7}{z^2} - \frac{3}{z^3}$ **B.** $-2z + 2 - \frac{6}{z} + \frac{1}{z^2} - \frac{3}{z^3}$

C. $-2z - 2 + \frac{8}{z} - \frac{7}{z^2} + \frac{3}{z^3}$ **D.** $4z - 2 - \frac{8}{z} - \frac{1}{z^2} + \frac{3}{z^3}$

Q6.14.1 MCQ 2014 2M

Let

$$X(z) = \frac{1}{1 - z^{-3}}$$

be the Z-transform of a causal signal $x[n]$. Then, the values of $x[2]$ and $x[3]$ are

A. 0 and 0 **B.** 0 and 1

C. 1 and 0 **D.** 1 and 1

Q6.14.2 MCQ 2014 2M

An input signal $x(t) = 2 + 5\sin(100\pi t)$ is sampled with a sampling frequency of 400 Hz and applied to the system whose transfer function is represented by

$$\frac{Y(z)}{X(z)} = \frac{1}{N}\left(\frac{1 - z^{-N}}{1 - z^{-1}}\right)$$

Where, N represents the number of samples per cycle. The output $y(n)$ of the system under steady state is

A. 0 **B.** 1 **C.** 2 **D.** 5

Q6.12.1 MCQ 2012 1M

If

$$x[n] = \left(\frac{1}{3}\right)^{|n|} - \left(\frac{1}{2}\right)^{n} u[n],$$

then the region of convergence (ROC) of its Z-transform in the Z-plane will be

A. $\frac{1}{3} < |z| < 3$ **B.** $\frac{1}{3} < |z| < \frac{1}{2}$

C. $\frac{1}{2} < |z| < 3$ **D.** $\frac{1}{3} < |z|$

Q6.09.1 MCQ 2009 2M

The z-transform of a signal $x[n]$ is given by

$$4z^{-3} + 3z^{-1} + 2 - 6z^{2} + 2z^{3}.$$

It is applied to a system, with a transfer function

$$H(z) = 3z^{-1} - 2.$$

Let the output be $y(n)$. Which of the following is true?

A. $y(n)$ is non causal with finite support.

B. $y(n)$ is causal with infinite support.

C. $y(n) = 0;\ |n| > 3.$

D. $Re\,[Y(z)]_{z=e^{j\theta}} = -Re\,[Y(z)]_{z=e^{-j\theta}}$;
$Im\,[Y(z)]_{z=e^{j\theta}} = Im\,[Y(z)]_{z=e^{-j\theta}}$, $-\pi \le \theta < \pi.$

Q6.08.1 MCQ 2008 2M

Given

$$X(z) = \frac{z}{(z-a)^2}$$

with $|z| > a$, the residue of

$$X(z)z^{n-1}$$

at $z = a$ for $n \ge 0$ will be

A. a^{n-1} **B.** a^{n} **C.** na^{n} **D.** na^{n-1}

Q6.08.2 MCQ 2008 2M

$H(z)$ is a transfer function of a real system. When a signal

$$x[n] = (1 + j)^{n}$$

is the input to such a system, the output is zero. Further the Region of Convergence (ROC) of

$$\left(1 - \frac{1}{2}z^{-1}\right)H(z)$$

is the entire Z-plane (except $z = 0$). It can then be inferred that $H(z)$ can have a minimum of

A. One pole and one zero

B. One pole and two zeros

C. Two poles and one zero

D. Two poles and two zeros

Q6.07.1 MCQ 2007 2M

$G(z) = \alpha z^{-1} + \beta z^{-3}$ is a low-pass digital filter with a phase characteristics same as that of the above question if

A. $\alpha = \beta$ **B.** $\alpha = -\beta$

C. $\alpha = \beta^{\frac{1}{3}}$ **D.** $\alpha = \beta^{-\frac{1}{3}}$

Q6.07.2 MCQ 2007 2M

A signal is processed by a causal filter with transfer function $G(s)$. For a distortion free output signal waveform, $G(s)$ must

A. Provide zero phase shifts for all frequency.

B. Provide constant phase shift for all frequency.

C. Provide linear phase shift that is proportional to frequency.

D. Provide a phase shift that is inversely proportional to frequency.

Q6.07.3 MCQ 2007 2M

$X(z) = 1 - 3z^{-1}$, $Y(z) = 1 + 2z^{-2}$ are Z-transforms of two signals $x[n]$, $y[n]$ respectively. A linear time invariant system has the impulse response $h[n]$ defined by these two signals as

$$h[n] = x[n-1] * y[n],$$

where $*$ denotes discrete time convolution. Then the output of the system for input $\delta[n-1]$

A. Has Z-transform $z^{-1}X(z)Y(z)$.

B. Equals $\delta[n-2]-3\delta[n-3]+2\delta[n-4]-6\delta[n-5]$.

C. Has Z-transform $1-3z^2+2z^{-2}-6z^{-3}$.

D. Does not satisfy any of the above three.

Q6.07.4 MCQ 2007 2M

Consider the discrete-time system shown in the figure where the impulse response of $G(z)$ is

$$g(0)=0,\quad g(1)=g(2)=1,\quad g(3)=g(4)=\cdots=0.$$

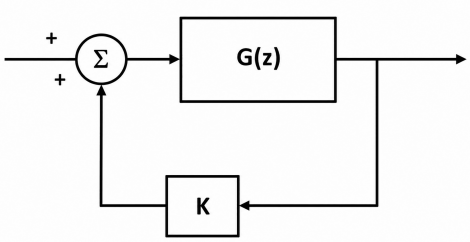


Fig. Q6.07.4

This system is stable for range of values of K

A. $\left[-1,\frac{1}{2}\right]$ **B.** $[-1,1]$ **C.** $\left[-\frac{1}{2},1\right]$ **D.** $\left[-\frac{1}{2},2\right]$

Q6.06.1 MCQ 2006 2M

$y[n]$ denotes the output and $x[n]$ denotes the input of a discrete-time system given by the difference equation

$$y[n]-0.8y[n-1]=x[n]+1.25x[n+1].$$

Its right-sided impulse response is

A. Causal **B.** Unbounded
C. Periodic **D.** Non-negative

Q6.06.2 MCQ 2006 2M

The discrete time signal

$$x[n]\ \leftrightarrow\ X(z)=\sum_{n=0}^{\infty}\frac{3^n}{2+n}z^{2n},$$

where $\leftrightarrow$ denotes a transform-pair relationship, is orthogonal to the signal.

A. $y_1[n]\ \leftrightarrow\ Y_1(z)=\sum_{n=0}^{\infty}\left(\frac{2}{3}\right)^n z^{-n}$

B. $y_2[n]\ \leftrightarrow\ Y_2(z)=\sum_{n=0}^{\infty}(5^{-n})\,z^{-(2n+1)}$

C. $y_3[n]\ \leftrightarrow\ Y_3(z)=\sum_{n=-\infty}^{\infty}2^{-|n|}z^{-n}$

D. $y_4[n]\ \leftrightarrow\ Y_4(z)=2z^{-4}+3z^{-2}+1$

Q6.05.1 MCQ 2005 2M

If $u(k)$ is the unit step and $\delta(k)$ is the unit impulse function, the inverse z-transform of

$$F(z)=\frac{1}{z+1}$$

for $k\geq 0$ is:

A. $(-1)^k\delta(k)$ **B.** $\delta(k)-(-1)^k$
C. $(-1)^k u(k)$ **D.** $u(k)-(-1)^k$

Answer Key

Q6.26.1 C	Q6.24.1 C	Q6.23.1 B
Q6.23.2 0.037 to 0.039	Q6.22.1 B	Q6.21.1 B
Q6.20.1 D	Q6.17.1 B	Q6.17.2 D
Q6.15.1 C	Q6.15.2 A	Q6.14.1 B
Q6.14.2 C	Q6.12.1 C	Q6.09.1 A
Q6.08.1 D	Q6.08.2 B	Q6.07.1 A
Q6.07.2 C	Q6.07.3 B	Q6.07.4 A
Q6.06.1 D	Q6.06.2 B	Q6.05.1 B

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